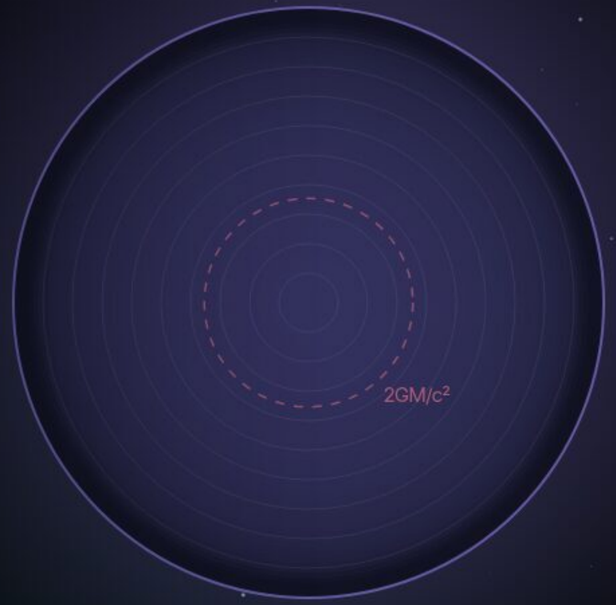
DEPARTMENT OF PHYSICS · **UDESC**

SEMINAR · **2026**

DARK MATTER IN NEUTRON STARS

# Simulating the *invisible*




What we know about dark matter, how a neutron star captures it *one collision at a time*, and what state the captured particles end up in.

$\mathscr{E} = \gamma\, e^{\nu(r)/2} < 1$ CAPTURED


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with Jorge A. Rueda · ICRANet and University of Ferrara


rafael-lima.pages.dev/talks

OUTLINE

# One problem, two papers, *one code*

## *I*

### The missing mass

How the problem was found, why the evidence holds, the ideas most explored today, and what the experiments have ruled out. Why look at a neutron star at all?

From Zwicky to the neutrino fog

## *II*

### OBSCURA: every collision

A relativistic Monte Carlo of capture that follows each particle through the interior metric, and reduces exactly to the optical-depth closure. What does the closure miss?

de Lima & Rueda, *A Monte Carlo relativistic framework for dark matter capture by neutron stars* (in preparation)

## *III*

### Degeneracy before self-gravity

After capture: a classical tracer, a degenerate Fermi gas, or a dark component? Two diagnostics, **S** for gravity and **Θ** for statistics, and a map of their thresholds.

de Lima & Rueda, *Degeneracy Before Self-Gravity: Dark Matter Regimes in Neutron Stars* (draft)

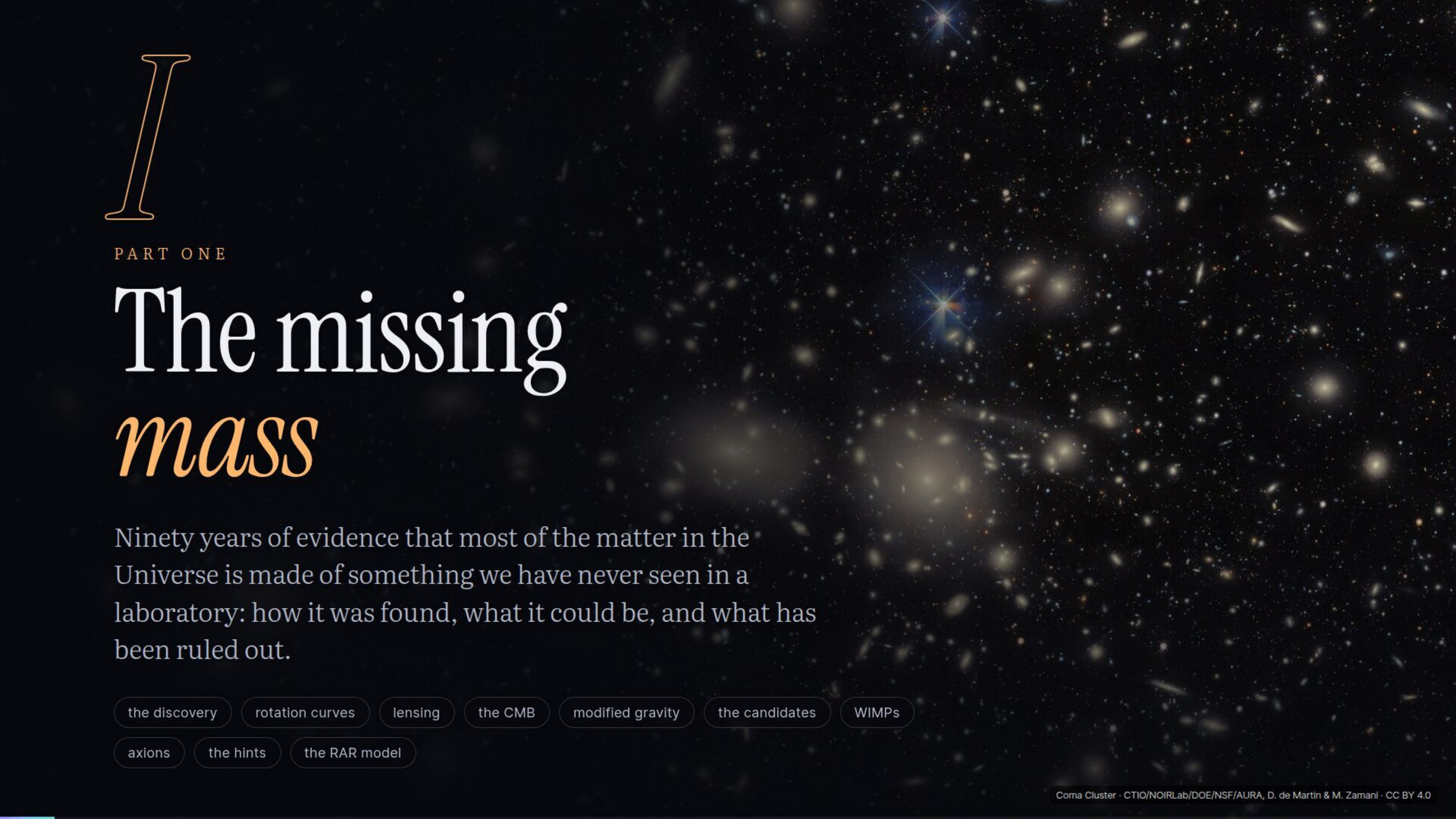
I

PART ONE

# The missing *mass*

Ninety years of evidence that most of the matter in the Universe is made of something we have never seen in a laboratory: how it was found, what it could be, and what has been ruled out.

the discovery · rotation curves · lensing · the CMB · modified gravity · the candidates · WIMPs · axions · the hints · the RAR model



Coma Cluster · CTIO/NOIRLab/DOE/NSF/AURA, D. de Martin & M. Zamani · CC BY 4.0

1933 · THE COMA CLUSTER

# Seven galaxies, *far too fast*




Fritz Zwicky

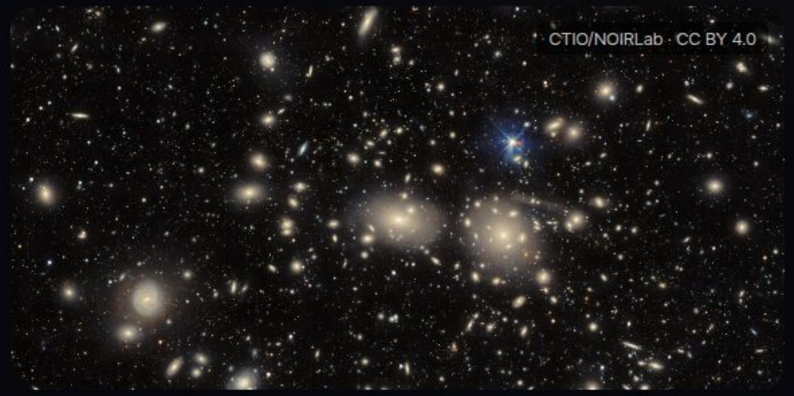
Helv. Phys. Acta 6, 110 (1933)

$$\langle v^2 \rangle = \frac{3}{5}\frac{GM}{R}$$

THE VIRIAL THEOREM, FOR A UNIFORM SPHERE

| | |
|---|---|
| 800 galaxies of $10^9$ M$\odot$ within $10^{24}$ cm | 80 km/s |
| Doppler spread he measured | ≳ 1000 km/s |
| density required | ≥ 400 × luminous |

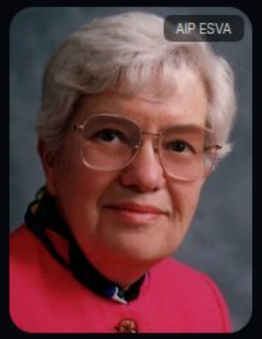
> *"...dark matter exists in much greater density than luminous matter."*
>
> Zwicky 1933, transl. H. Andernach (2017)




**WHAT SURVIVED**

Today's $H_0$ is seven times smaller than his, and X-rays later found hot gas outweighing the galaxies. The excess shrinks and stays: baryons are about **one sixth** of a cluster.

# Galaxies that *refuse to slow down*







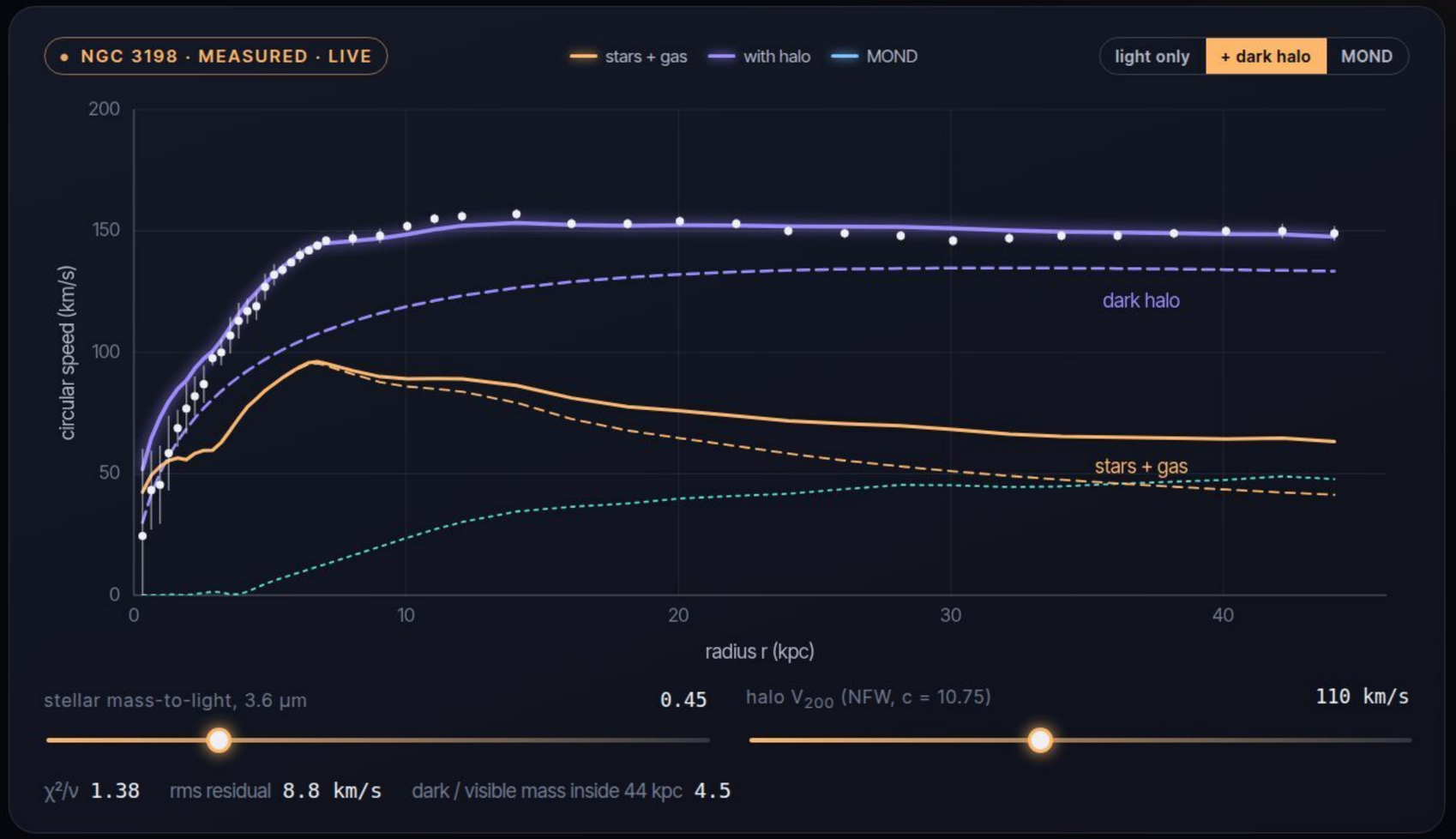
## Vera Rubin

with Kent Ford, optical spectra from 1970; Albert Bosma in 21 cm, beyond the stars.

$$v^2(r) = \frac{GM(<r)}{r}$$

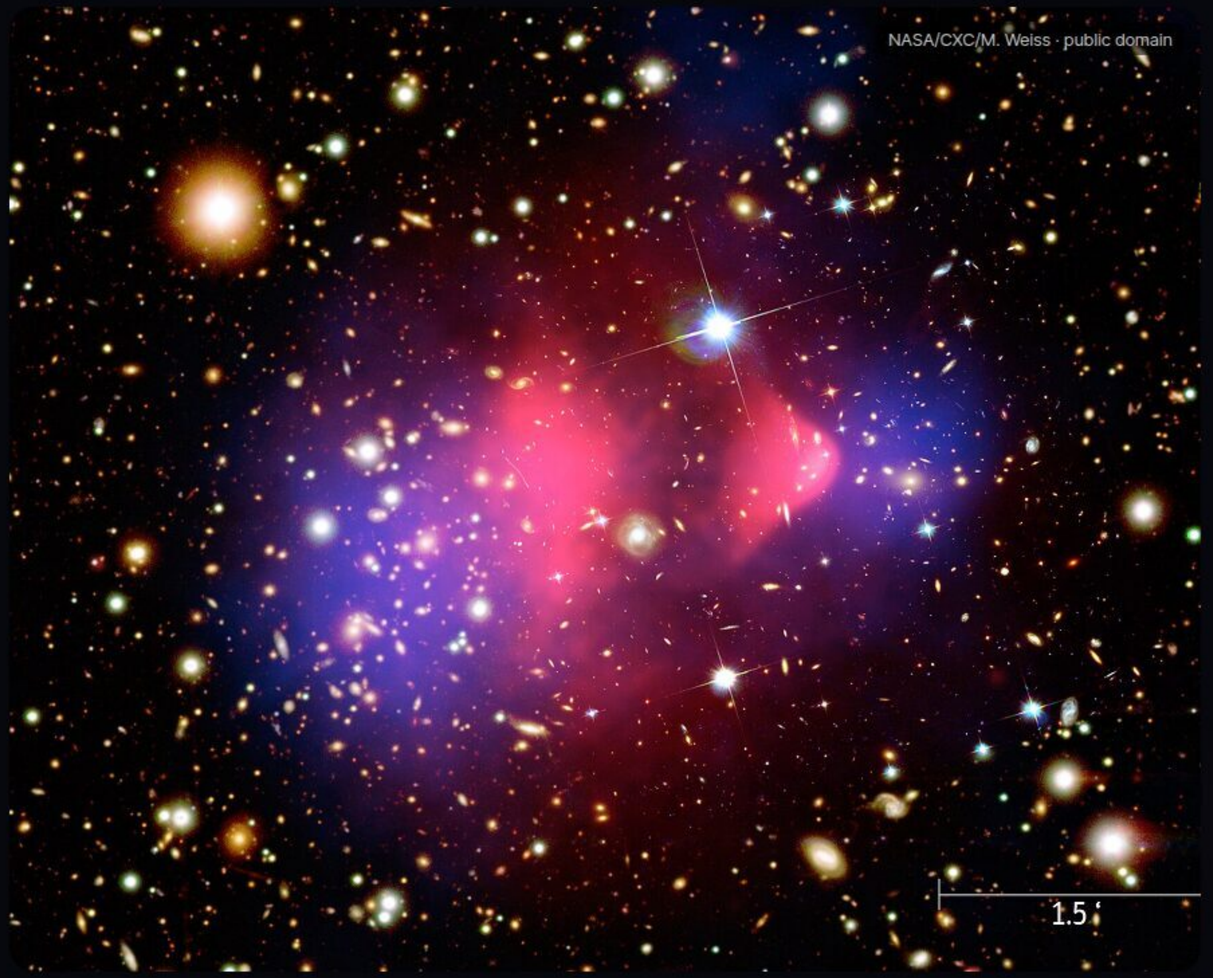
FLAT v MEANS M ∝ r

Rubin & Ford 1970; Rubin, Ford & Thonnard 1978, 1980; Bosma 1978. Data: SPARC, Lelli, McGaugh & Schombert, AJ 152, 157 (2016).



2006 · THE BULLET CLUSTER

# Mass that *left the gas behind*




$$\theta_E^2 = \frac{4GM}{c^2}\,\frac{D_{ls}}{D_l D_s}$$

LENSING WEIGHS MASS, LUMINOUS OR NOT

**Two clusters went through each other.** The gas (pink, X-rays), which holds most of the baryons, was braked. The lensing mass (blue) kept going with the galaxies.

**8σ OFFSET**

Between the total mass and the baryons, and it “cannot be explained with an alteration of the gravitational force law.” Clowe et al. 2006

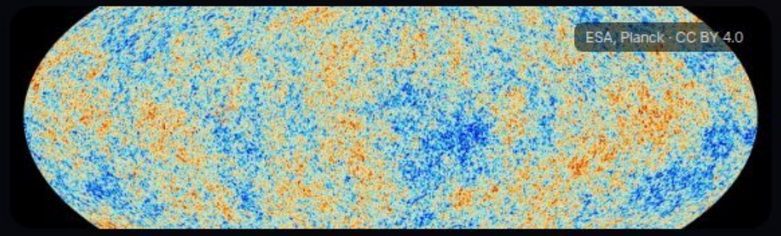
**72 COLLISIONS LATER**

Dark matter stays with the stars: $\sigma/m < 0.47\ \mathrm{cm^2/g}$ at 95%. It barely collides, even with itself. Harvey et al. 2015

380 000 YEARS AFTER THE BIG BANG

# The early Universe *weighs it*




**Baryons** load the photon fluid: compression peaks rise over rarefaction peaks.

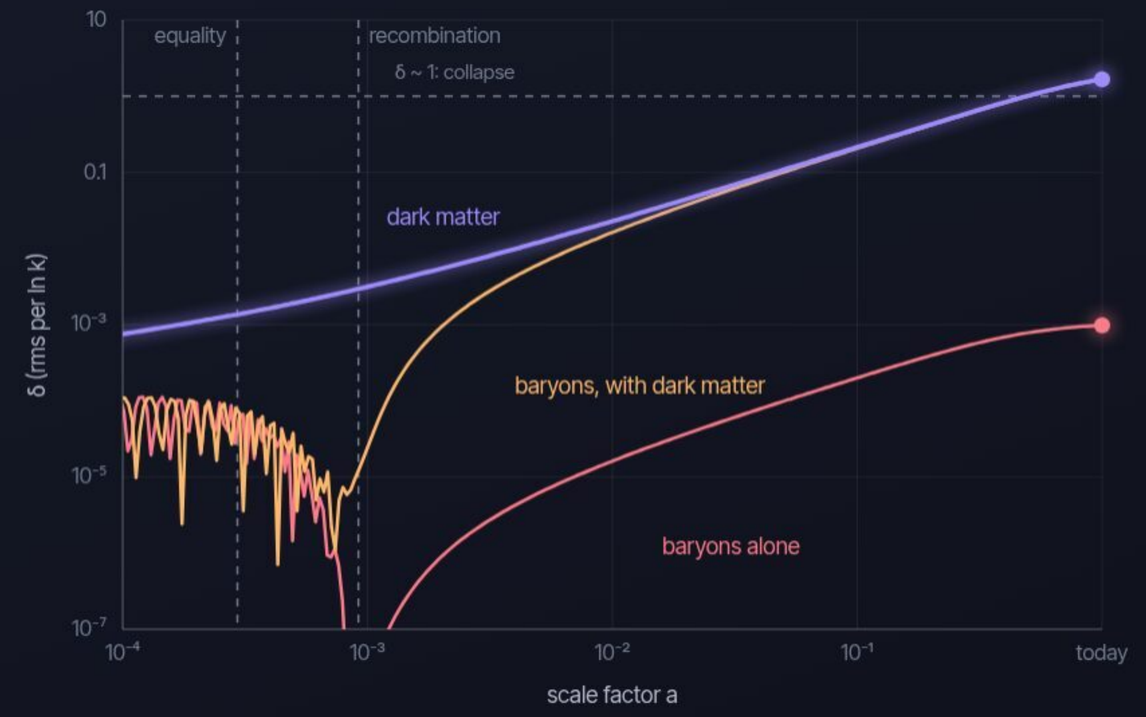
**Dark matter** feels no pressure, so its wells do not decay. Without it, the first peak more than doubles.

$$\Omega_c h^2 = 0.120 \pm 0.001$$
$$\Omega_b h^2 = 0.0224 \pm 0.0001$$

PLANCK 2018

THREE MINUTES · 380 000 YEARS · TODAY

# It cannot be *atoms*




Same primordial fluctuations. With dark matter, $\delta \approx 1.7$ today; with baryons alone, $\delta \approx 10^{-3}$: no galaxies.

### TWO CLOCKS AGREE ON THE BARYONS

Primordial deuterium, $(D/H)_p = (2.527 \pm 0.030) \times 10^{-5}$, fixes the baryon density at three minutes. It matches the CMB, 380 000 years later, within $2\sigma$.




atoms 4.9%, so dark matter is about 85% of all matter · Planck 2018, h = 0.674

### NOT NEUTRINOS EITHER

$\Sigma m_\nu < 0.12$ eV makes them at most about **1%** of the dark matter, and too hot. **Nothing in the Standard Model fits.**

THE ALTERNATIVE

# Or is *gravity* wrong?

$$a = (a_N\, a_0)^{1/2} \text{ for } a_N \ll a_0$$
$$v^4 = GM\, a_0$$

MOND, MILGROM 1983: $a_0 \approx 1.2\times10^{-10}$ m/s²

The regularity is real: in rotating galaxies, the baryons alone predict the acceleration. The MOND button in the rotation-curve panel uses exactly that relation, with no halo.

Any dark-matter theory has to **explain** it. But a theory of gravity has to work on **every** scale.

| SCALE | TEST | |
|---|---|---|
| **Galaxies** | 2693 points in 153 galaxies follow one relation with one constant | WORKS |
| **Wide binaries** | Gaia DR3: Newton preferred at 19σ (Banik et al.), an anomaly at low *g* (Chae) | DISPUTED |
| **Clusters** | Still short of mass by about a factor of 2, and in the Bullet the potential follows the galaxies | FAILS |
| **GW170817** | Gravity and light arrive together: the end of "dark-matter emulators" | FAILS |
| **CMB, *P*(*k*)** | AeST fits both, with a new field that behaves as dust in the early Universe | AT A PRICE |

# We know how much, *not what*

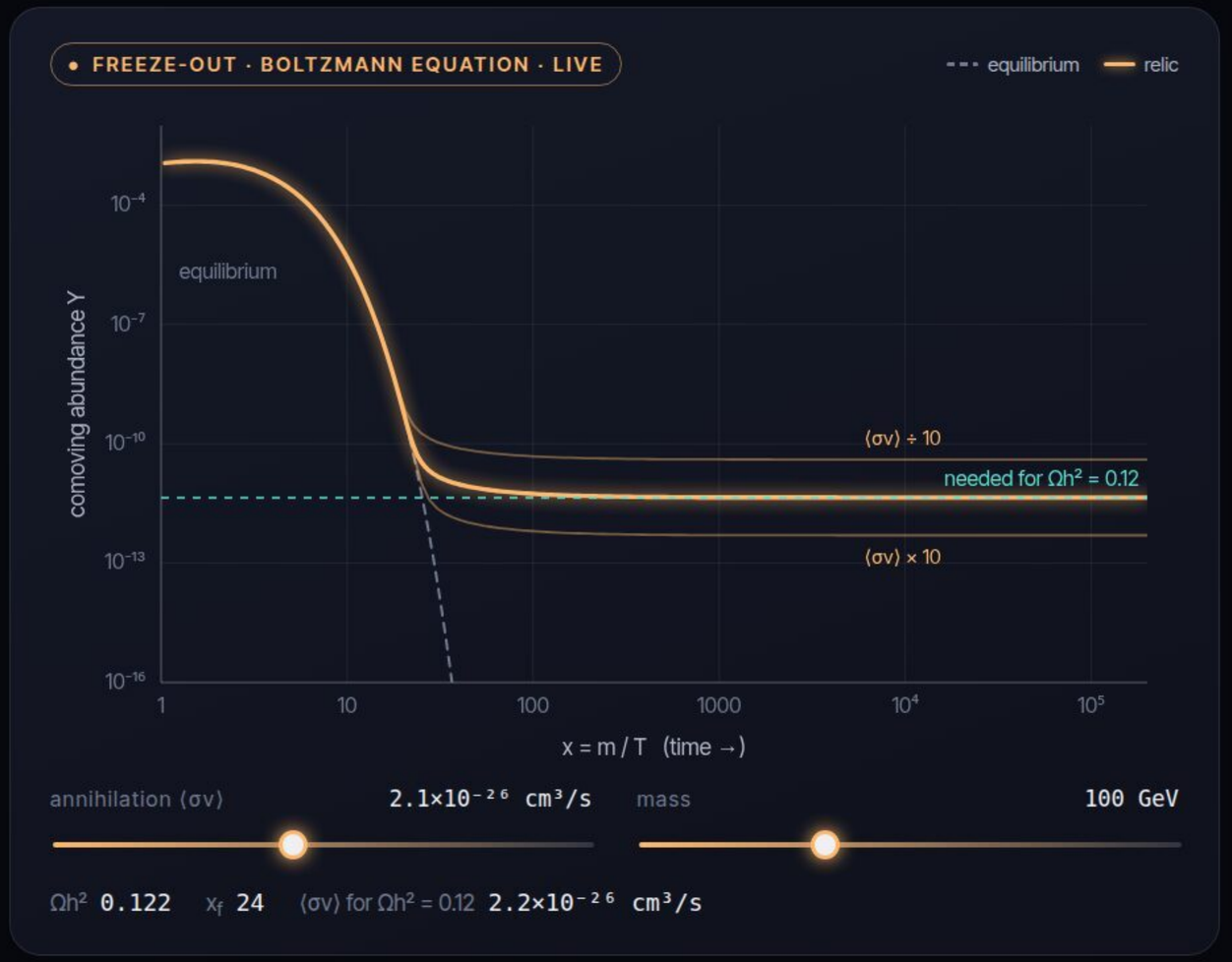
non-baryonic · cold · neutral · stable over cosmic time · nearly collisionless · $\Omega_c h^2 = 0.120$ · $\rho_\chi \approx 0.4$ GeV/cm$^3$ near the Sun




Bounds: ultralight scalars $m > 2\times10^{-20}$ eV (Rogers & Peiris 2021) · Pauli, any fermion $m \geq 0.13$ keV (Alvey et al. 2021) · thermal warm dark matter $m > 5.7$ keV (Iršič et al. 2024) · compact objects $10^{-8}$–$10^{3}$ M⊙ not the dominant component (OGLE, Mróz et al. 2024, 2025) · asteroid-mass window open (Gorton & Green 2024). Bright bands: the most explored today.

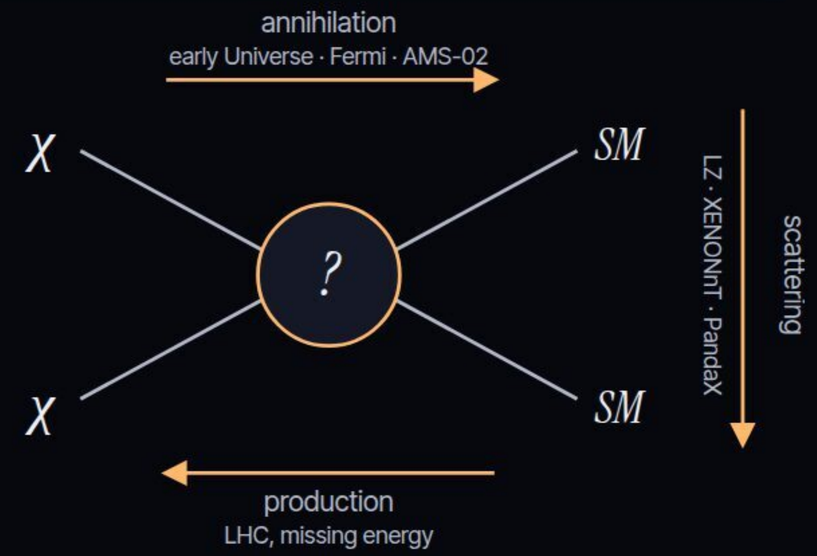
THE MOST EXPLORED IDEA

# The *WIMP* miracle



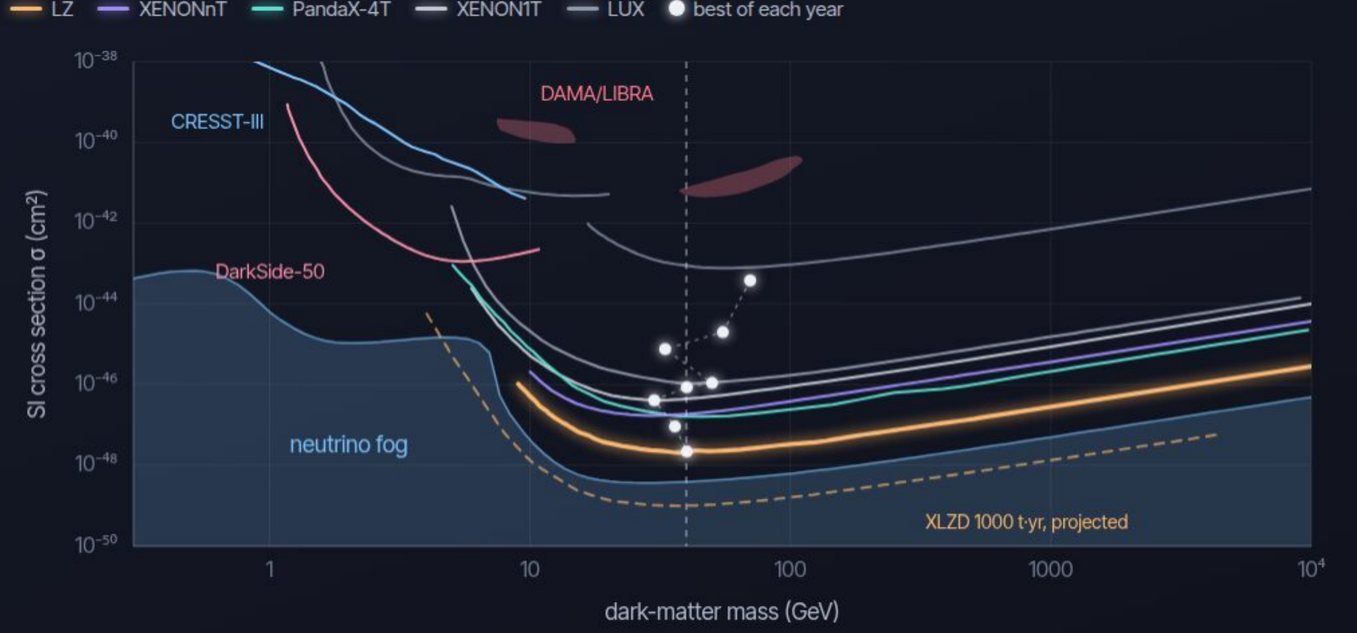

$$\frac{dY}{dx} = -\frac{\lambda}{x^2}(Y^2 - Y_{eq}^2)$$

THE RELIC DOES NOT DEPEND ON THE MASS, ALMOST




$\Omega h^2 = 0.12$ needs $\langle\sigma v\rangle \approx 2.2\times10^{-26}$ cm³/s above 10 GeV (Steigman, Dasgupta & Beacom 2012).

# Four orders of magnitude, *no WIMP*




$$\frac{dR}{dE_R} \propto \frac{\rho_\chi \sigma A^2 F^2(E_R)}{m_\chi \mu^2} \int \frac{f(v)}{v} d^3v$$

A²: HENCE TONNES OF XENON

### THE FOG HAS ARRIVED

LZ, 5.7 t·yr: **$^8$B solar neutrinos** scattering coherently on xenon, at **4.5σ**. The next limits must separate them from a signal.

### THE BLIND SPOT

Recoils fall as $\mu^2 v^2 / m_A$. Below a few GeV they drop under threshold: at 100 keV, about **1 μeV** on xenon.

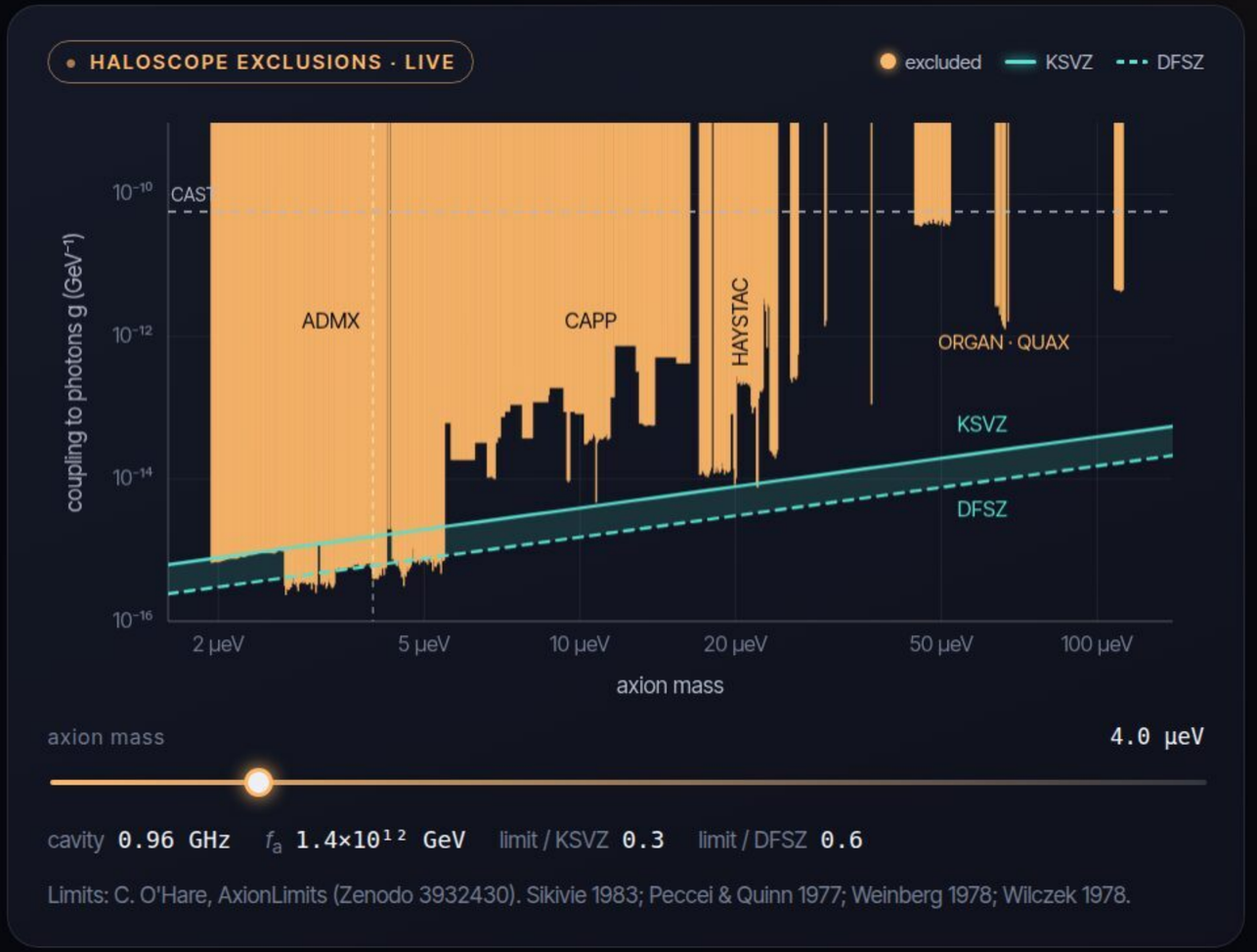
Limits: compiled by C. O'Hare. Fog: O'Hare, PRL 127, 251802 (2021). Best points: the papers' abstracts.

# A particle invented *for another problem*

1. QCD allows a CP-violating angle $\bar{\theta}$, yet the neutron's dipole moment gives $|\bar{\theta}| \lesssim 10^{-10}$.
2. Peccei–Quinn: make $\bar{\theta}$ a field that relaxes to zero. Its quantum is the **axion**.
3. Oscillating coherently since the QCD epoch, it behaves as **cold dark matter**.
4. In a magnetic field $a \to \gamma$: a tuned cavity collects photons of $h\nu = m_a c^2$.

$$m_a \approx 5.7\ \mu\text{eV}\ \frac{10^{12}\ \text{GeV}}{f_a}$$

Haloscopes reach the benchmark **DFSZ** line only around **3–5 μeV**, a tenth of the range scanned.




Limits: C. O'Hare, AxionLimits (Zenodo 3932430). Sikivie 1983; Peccei & Quinn 1977; Weinberg 1978; Wilczek 1978.

# Hints, and *what became of them*

| CLAIM | SINCE | WHERE IT STANDS | |
|---|---|---|---|
| **DAMA/LIBRA** annual modulation | 1998 | Same NaI target, no modulation: COSINE-100 + ANAIS-112 exclude it at 4.7σ (1–6 keV) and 3.5σ (2–6 keV) | REFUTED |
| **XENON1T** electron excess | 2020 | A 3.4σ solar-axion fit; gone in XENONnT, with a background five times lower | GONE |
| **3.5 keV** X-ray line | 2014 | XRISM sees no line in ten clusters; its limit is still five times above the XMM detection | FADING |
| **Galactic Centre** γ-ray excess | 2009 | The photons are real; recent work leans to unresolved millisecond pulsars, but it is not settled | OPEN |
| **20 GeV** Milky Way halo excess | 2025 | Reproduced in Fermi data; in tension with dwarf galaxies and, in a preprint, with AMS-02 antiprotons | OPEN |
| **LHC** missing energy | Run 1–3 | No established excess; no new particle at the weak scale so far | NOTHING |

The cleanest indirect targets stay dark too: dwarf galaxies put $\langle\sigma v\rangle$ **below the thermal value** for $m \lesssim 100$ GeV in quark and τ channels (Fermi-LAT 2015). Status as of November 2026.

# What if dark matter is *a Fermi gas?*

Take neutral fermions, gravity and the Pauli principle, and solve Einstein's equations at finite temperature. *Nothing else.*

**1939** — **Oppenheimer & Volkoff**
Pauli against gravity: a limit mass

**1969** — **Ruffini & Bonazzola**
self-gravitating quantum particles

**1983–92** — **Ruffini & Stella; Ingrosso et al.**
semi-degenerate, energy cutoff

**2002** — **Bilić et al.**
a Newtonian fermion ball at the Galactic Centre

**2015** — **Ruffini, Argüelles & Rueda**
core, plateau and halo in GR

**2018** — **Argüelles et al.**
the cutoff: a core at Sgr A*, a halo for the Galaxy

**ONE PARTICLE**

One distribution function and one fermion mass, from milliparsecs to a hundred kiloparsecs.

**NO NEW FORCE**

The pressure comes from quantum statistics and temperature. No interaction beyond gravity is needed.

**THE MASS SHOWS**

Unlike cold dark matter, the particle mass sets the structure: a degenerate core shrinks as $m^{-8/3}$.

# Fermi–Dirac, *with a way out*

$$f_c(\varepsilon \le \varepsilon_c) = \frac{1 - e^{(\varepsilon - \varepsilon_c)/kT}}{e^{(\varepsilon - \mu)/kT} + 1}, \quad f_c(\varepsilon > \varepsilon_c) = 0$$

FERMI–DIRAC, EMPTIED ABOVE THE ESCAPE ENERGY

$$\rho = m \frac{2}{h^3} \int f_c \, (1 + \varepsilon/mc^2) \, d^3p, \quad P = \frac{1}{3} \frac{2}{h^3} \int f_c \, p \, v \, d^3p$$

THE EQUATION OF STATE, AT EVERY RADIUS; TOV CLOSES THE SYSTEM

Not a guess: a maximum-entropy state of collisionless relaxation with evaporation.

Lynden-Bell 1967; Chavanis 2004; Ruffini, Argüelles & Rueda 2015; Argüelles et al., PDU 21, 82 (2018).

**TOLMAN**

$e^{\nu/2}\,T$ = const: hotter deeper in the well

**KLEIN**

$e^{\nu/2}\,(\mu + mc^2)$ = const: more degenerate deeper in

**ENERGY ALONG A GEODESIC**

$e^{\nu/2}\,(\varepsilon_c + mc^2)$ = const: one conserved escape energy

$$\beta = \frac{kT}{mc^2}, \; \theta = \frac{\mu}{kT}, \; W = \frac{\varepsilon_c}{kT} \Rightarrow W(r) = W_0 + \theta(r) - \theta_0$$

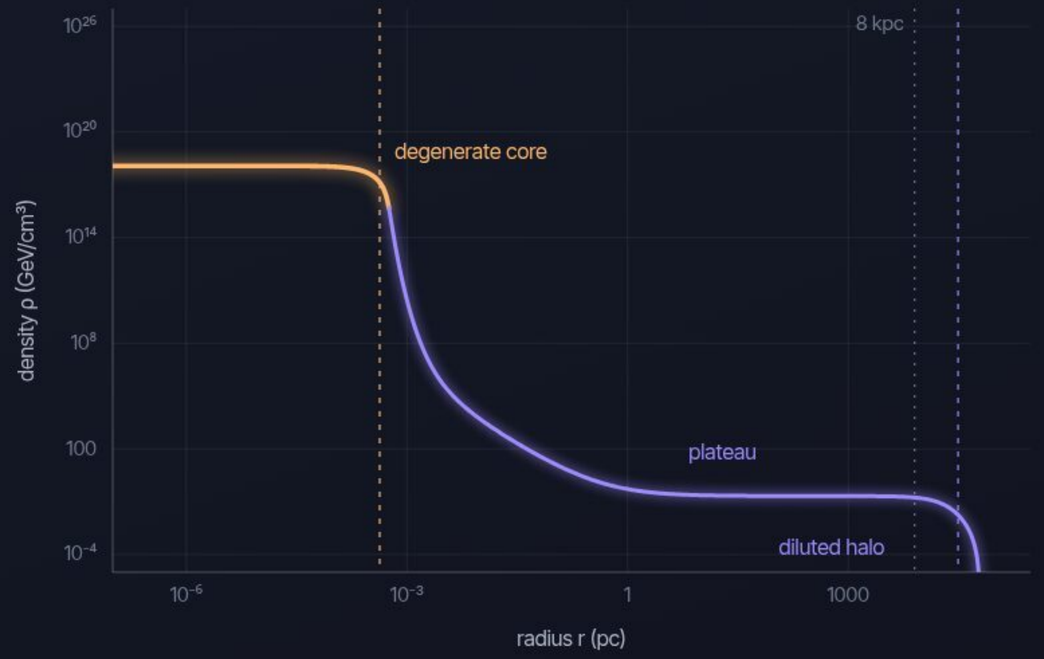
FREE: $m$, $\beta_0$, $\theta_0$, $W_0$

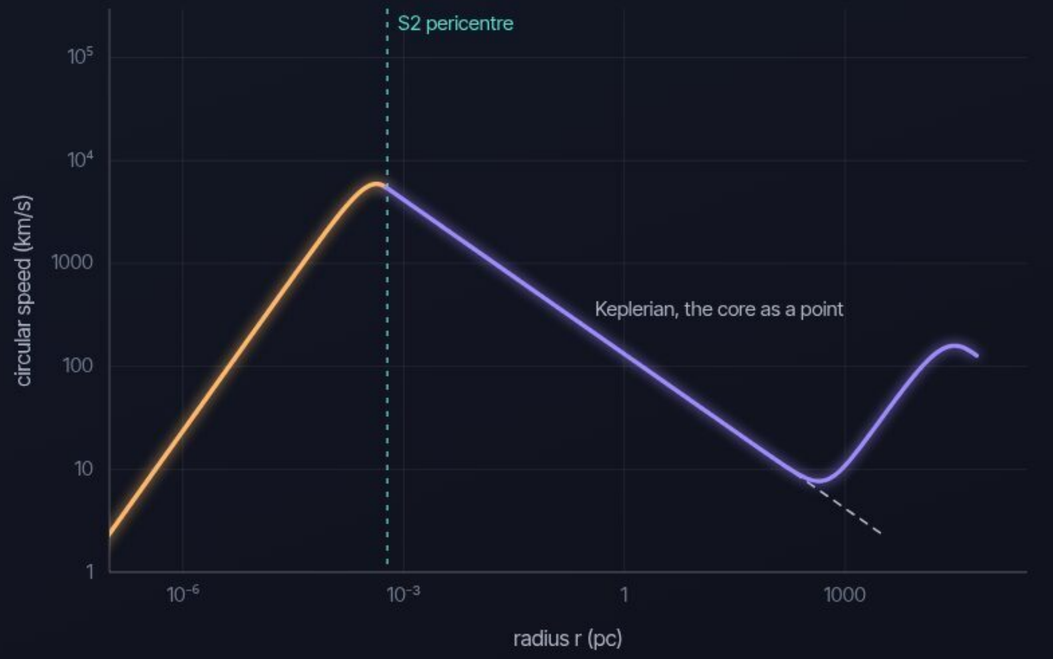
# One gas, *three regimes*

• TOV + FERMI–DIRAC · SOLVED IN YOUR BROWSER

FERMION, KEV: 48 | 56 | 100 | 345

RIGHT: rotation | θ and W | f(v) at 8 kpc



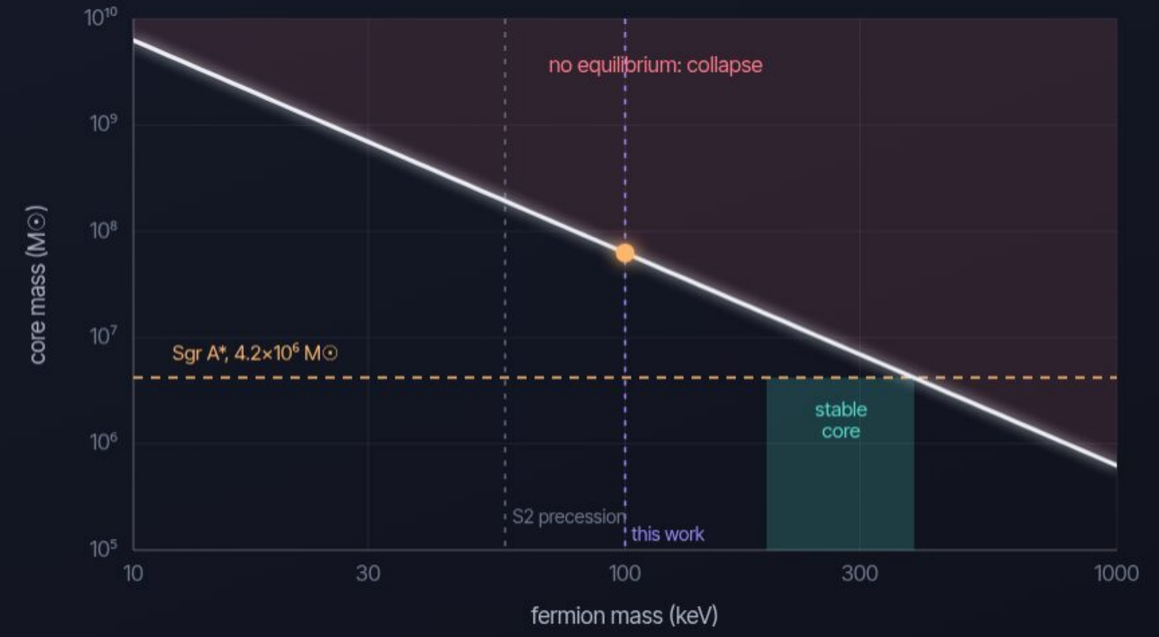

central degeneracy $\theta_0$ 37.77

central temperature $\beta_0$ 1.20×10⁻⁵

halo escape speed, c√(2Wβ) 282 km/s

core 3.5×10⁶ M⊙ in 0.427 mpc | seen from outside 4.05×10⁶ M⊙ | halo peak 159 km/s at 31.2 kpc | edge 62.0 kpc | total 2.3×10¹¹ M⊙ | ρ(8 kpc) 0.17 GeV/cm³ | $W_0$ 66.341

# A neutron star, *rescaled*




fermion mass **100 keV**

limit **6.26×10⁷ M⊙** its radius **5.61 AU** central density **0.46 g/cm³**

a Sgr A* ball, non-relativistic **0.12 mpc** (S2 pericentre 0.6)

$$M_{\mathrm{OV}} = 0.384\,\frac{m_{\mathrm{P}}^{3}}{m^{2}}$$

ONE FORMULA, ONE NUMBER FOR EACH FERMION

| IDEAL FERMIONS | NEUTRONS | 100 KEV |
|---|---|---|
| Particle | 939.6 MeV | 100 keV |
| Limit mass | 0.71 M⊙ | 6.3×10⁷ M⊙ |
| Radius there | 9.5 km | 5.6 AU |
| Central density | 3.6×10¹⁵ g/cm³ | 0.46 g/cm³ |
| Radius / $R_S$ | 4.5 | 4.5 |

Lighter than water, and at the edge of collapse. Past the limit, a **black-hole seed** with no star involved.

Argüelles et al. 2018; Krut, Argüelles & Chavanis 2025; Argüelles et al. 2024.

# A black hole, *or a core?*

| TEST | RESULT | |
|---|---|---|
| **Rotation curve** | One continuous profile, with baryons, from the core to the Gaia DR3 curve | WORKS |
| **S2 and G-source orbits** | Core and black hole differ by less than 1% in orbital parameters; no conclusive preference | UNDECIDED |
| **S2 precession** | Retrograde below 56 keV, prograde above. GRAVITY's prograde signal points to $m \gtrsim 57$ keV, with under 0.1% of the core inside the orbit | BOUNDS m |
| **Stability** | Stable core–halo solutions with a $4.2\times10^6$ M☉ core exist only for **194–387 keV** | BOUNDS m |
| **Images** | Central depression and a ring, down to about 35 μas, but **no photon ring**: the key difference from a black hole | NEXT TEST |
| **Collapse** | Cores past the limit become black-hole seeds, sooner if baryons accrete onto them | PREDICTION |

Our **100 keV** is a capture benchmark, not a claim about Sgr A*: it sits inside the 48–345 keV of the halo fits, below the stable window for a core that replaces the black hole. Status as of November 2026.

Argüelles et al. 2018; Becerra-Vergara et al. 2020, 2021; Argüelles et al. 2022; Pelle et al. 2024; Krut, Argüelles & Chavanis 2025; Crespi et al. 2026.

THE PREMISE

# A neutron star is a *dark-matter detector*

## As it crosses the Galaxy, a neutron star sweeps up dark matter: its potential well *focuses* the particles, its matter *stops* them.

halo → focusing → collisions → **capture** → accumulation → regime → heating · collapse · structure

### A DEEP WELL

Escape speed at the surface $\sqrt{(2GM/R)}$ = **0.59 c** for 1.4 $M_\odot$ and 11.7 km. Slow halo particles are bent toward the star from a thousand radii away.

### A DENSE TARGET

$n_b \approx 5\times10^{38}$ cm$^{-3}$ at the centre: with $\sigma = 10^{-44}$ cm$^2$ a particle meets $\tau \approx$ **6–9** collisions in one crossing.

### AN OLD, COLD STAR

What accumulates can heat it, collapse into a black hole, or change its structure — **if** we know how much is captured and in what state.

II

PART TWO

# OBSCURA: *every collision*

A relativistic Monte Carlo that follows each dark-matter particle through the interior metric of the star, collision by collision — and recovers the optical-depth closure when its extra physics is switched off.

the halo · focusing · the optical-depth closure · the transport · what the closure misses · count laws · cooling



Bullet Cluster, 1E 0657–56 · NASA/CXC/M. Weiss · public domain

# Two ways to fill *the halo*

### STANDARD HALO · MAXWELL–BOLTZMANN

$$f(v) \propto v^2 e^{-v^2/v_0^2},\ v < v_{\mathrm{esc}}$$

$v_0$ = 220 km/s, $v_{\mathrm{esc}}$ = 544 km/s

The usual WIMP benchmark: particles of GeV to TeV, a few hundred km/s, isotropic. Used for the Maxwell–Boltzmann results of Part III.

### RAR FERMIONS · TRUNCATED FERMI–DIRAC

$$f \propto \frac{1 - e^{(E - E_c)/kT}}{e^{(E-\mu)/kT} + 1},\ E \le E_c$$

RUFFINI–ARGÜELLES–RUEDA

Self-gravitating fermions in general-relativistic equilibrium: a degenerate core at the Galactic centre and a Boltzmann halo. Rotation curves and the S-stars allow **48–345 keV**; we take **100 keV**.



FORTY YEARS OF CAPTURE

# From the Sun to *neutron stars*

**1985**
**Press & Spergel**
capture of WIMPs by the Sun

**1987**
**Gould**
the capture-rate formalism

**1989**
**Goldman & Nussinov**
WIMPs and neutron stars

**2008–10**
**Kouvaris; Kouvaris & Tinyakov**
heating by annihilation; collapse to a black hole

**2017**
**Baryakhtar et al.**
dark kinetic heating

**2020**
**Bell et al.**
relativistic capture with Pauli blocking, opacity, multiple scattering

**2026**
**OBSCURA**
collision histories through the interior metric

# The orbit must *reach the surface*

1. Outside the star, two constants: $\mathscr{E} = (1 - 2M/r)\,u^t$, $\mathscr{L} = r^2 u^\varphi$. From infinity, $\mathscr{E} = \gamma$ and $\mathscr{L} = \gamma v b$.
2. $u_\mu u^\mu = -1$ leaves one dimension: $(u^r)^2 = \mathscr{E}^2 - V^2$, with $V^2 = (1 - 2M/r)(1 + \mathscr{L}^2/r^2)$.
3. No turning point before the surface: $\mathscr{E}^2 \geq V^2(R)$, with $C = M/R$.
4. That caps the angular momentum: $\mathscr{L}_c = R\,[(\mathscr{E}^2 - 1 + 2C)/(1 - 2C)]^{1/2}$.

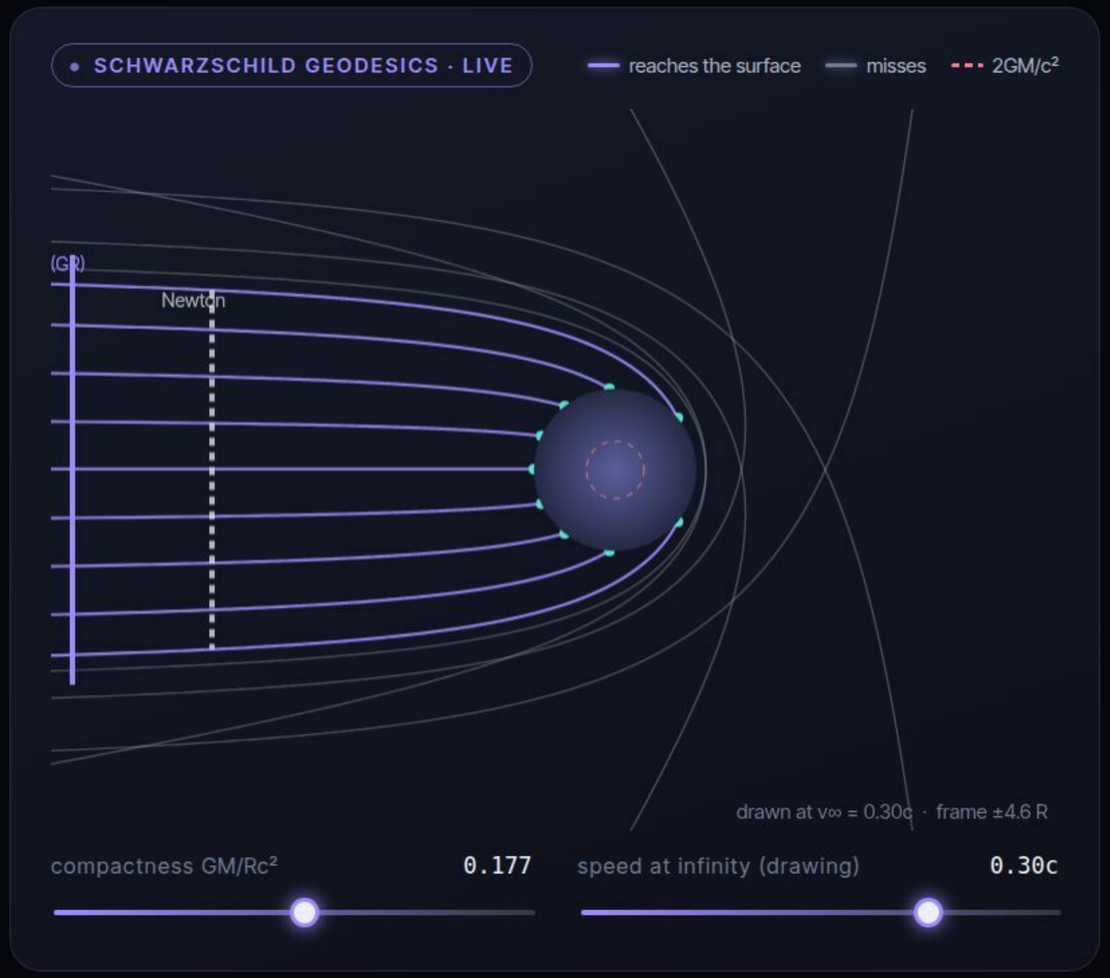
$$\frac{b_{max}^2}{R^2} = \frac{1}{1-2C}\left[1 + \frac{2C}{(\gamma v)^2}\right] \text{ vs Newton } 1 + \frac{2C}{v^2}$$

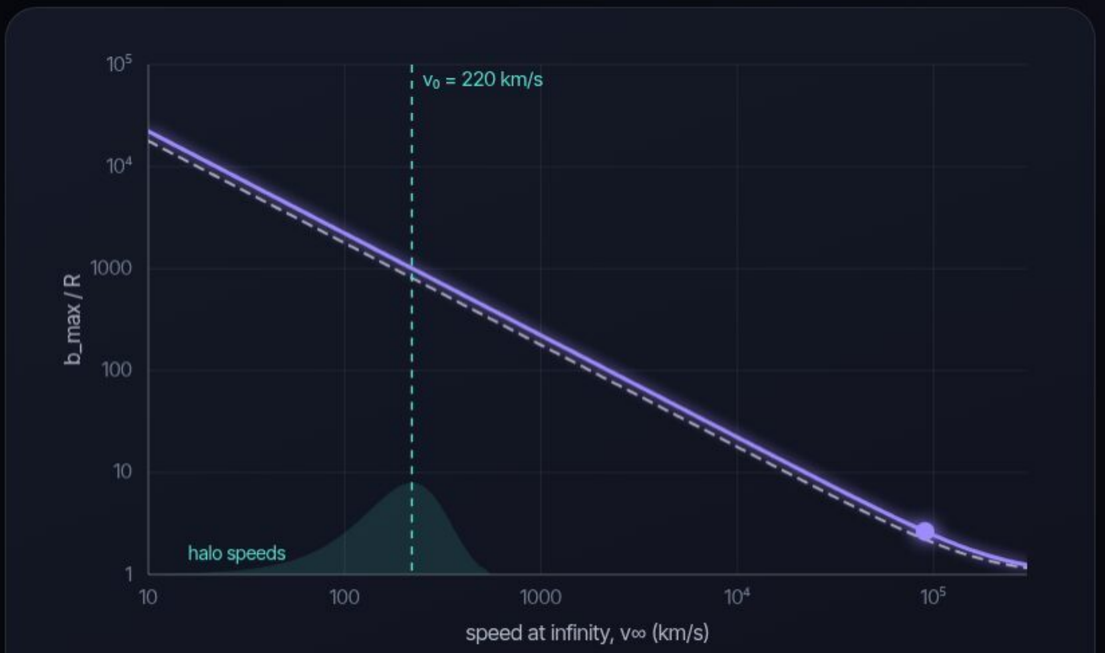
TWO CORRECTIONS: v → γv, AND THE APPARENT RADIUS R/√(1 − 2C)

Exact for C ≤ 1/4. Above that, slow particles meet the barrier **outside** the star and $\mathscr{L}_c \to 4M$.



# The star looks *a thousand times wider*




$$\frac{b_{\max}}{R} = \frac{1}{\gamma v}\sqrt{(\gamma^2 - 1 + 2C)\,/\,(1 - 2C)}$$

FROM $\mathscr{L} < \mathscr{L}_c(\mathscr{E})$: THE ORBIT MUST REACH R

at this speed `b_max = 2.65 R` · Newton `2.21 R` · GR/Newton `1.198`
· at 220 km/s: `1010 R` , × `1.244` · slow limit 1/√(1−2C) = `1.244`

SUPPLY × PROBABILITY

# What reaches the star, and *what stays*

$$ds^2 = -e^{\nu(r)}dt^2 + e^{\lambda(r)}dr^2 + r^2 d\Omega^2$$

THE TOV STAR: SLY, 1.40 M⊙, 11.70 km, C = 0.177

$$\frac{d^2\dot{N}_{\rm in}}{d\mathscr{E}\, d\mathscr{L}} = 8\pi^2\, m^3\, f(\mathscr{E})\, \mathscr{L}, \quad \mathscr{L} < \mathscr{L}_c(\mathscr{E})$$

PARTICLES ENTERING THE SURFACE, PER UNIT COORDINATE TIME

$$\dot{N}_{\rm X} = 8\pi^2 m^3 \iint P_{\rm cap}(\mathscr{E}, \mathscr{L})\, f(\mathscr{E})\, \mathscr{L}\, d\mathscr{L}\, d\mathscr{E}$$

THE CAPTURE RATE: THE KERNEL $P_{cap}$ IS WHAT OBSCURA COMPUTES

$\mathscr{E}$ and $\mathscr{L}$ are the energy and angular momentum per unit mass, measured at infinity; between collisions they are **conserved**, and the particle follows the interior geodesic.

**CAPTURED**

$$\mathscr{E}_{\rm final} = \gamma_{\rm final}\, e^{\nu(r_{\rm coll})/2} < 1$$

after a collision the particle can no longer reach infinity: $\gamma$ is its Lorentz factor for a static observer at the collision, $e^{\nu/2}$ the lapse there.

THE OPTICAL-DEPTH CLOSURE

# A Poisson count and *three frozen choices*

$$P_{\mathrm{cap}} = 1 - e^{-\tau} \sum_{k < N_{\mathrm{min}}} \frac{\tau^k}{k!}$$

THE NUMBER OF COLLISIONS IS POISSON WITH MEAN T

$$\tau = 2 \int_{r_{\mathrm{tp}}}^{R} \frac{e^{(\nu+\lambda)/2}\, \Gamma_{n\chi}}{\sqrt{(\mathcal{E}^2 - V_{\mathrm{eff}}^2)}}\, dr$$

COLLISIONS ALONG THE INCIDENT GEODESIC, ONE CROSSING

The closure of Sirvente & Rueda (submitted): relativistic and covariant. OBSCURA was built to test it, not to replace it — and reduces to it exactly.

**1 · FROZEN TRAJECTORY**

Collisions change the energy, never the path: every particle keeps the incident geodesic.

**2 · DETERMINISTIC LOSS**

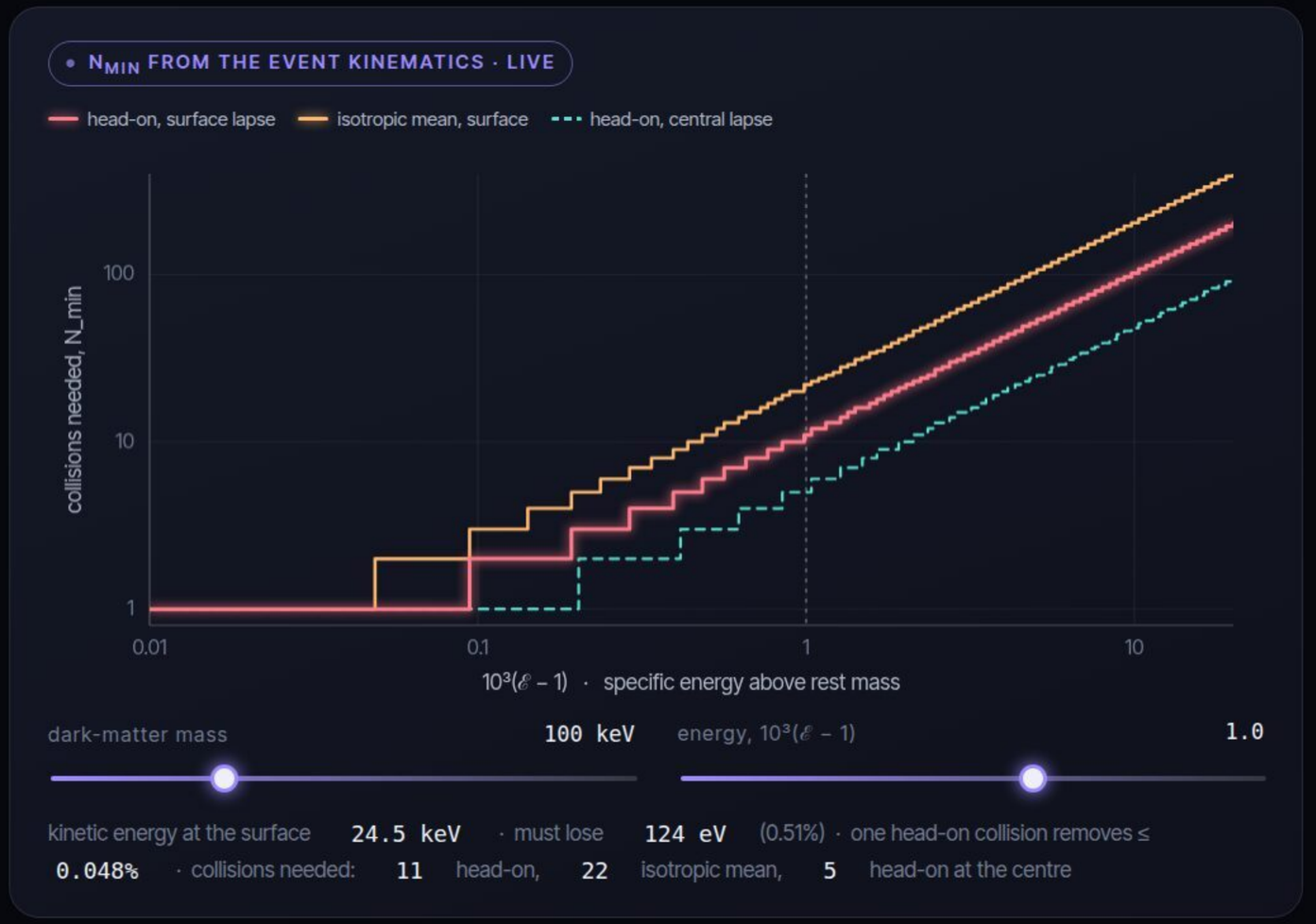
Each collision removes a fixed amount — head-on or the isotropic mean — so capture needs exactly $N_{\mathrm{min}}$ of them.

**3 · EVALUATED AT THE SURFACE**

$N_{\mathrm{min}}$ is computed with the lapse at R, wherever the collision actually happens.

100 KEV ON A NEUTRON OF 939 MEV

# Capture decided in *a narrow window*




$$\frac{T}{K} = \frac{m_n m_\chi (\gamma + 1)(1 - \cos\theta^*)}{s}$$

KINETIC ENERGY GIVEN TO THE NEUTRON, EXACT

For $m_\chi \ll m_n$ each collision removes a fraction $\sim 2(m_\chi/m_n)(\gamma + 1) \approx 5\times10^{-4}$ at most, while it must lose up to a per cent of it: tens of collisions.

Deeper in the star the lapse is smaller and the same collision removes more of $\mathscr{E}$: $N_{min}$ **halves** at the centre.

Slide the mass to the GeV range: one collision suffices — the WIMP limit.

THE TRANSPORT

# One history, *collision by collision*

1. Draw a target optical depth $-\ln U$ and fly along the **interior geodesic** with $(\mathscr{E}, \mathscr{L})$ conserved, accumulating $d\tau = n_b \sigma\, dl$ (step R/4096).
2. At the collision, an **elastic two-body event** on a neutron at rest, isotropic in the centre of mass: energy loss and deflection from the same angles.
3. New $\gamma$ and direction → new $(\mathscr{E}, \mathscr{L})$: the **orbit is rebuilt**.
4. **Captured** if $\mathscr{E} < 1$; **escaped** if it leaves unbound; otherwise fly again.

$$T = \frac{p^{*2}}{m_n}(1 - \cos\theta^*),\ p^* = \frac{m_n p_\chi}{\sqrt{s}}$$

EXACT, FROM THE INVARIANT t

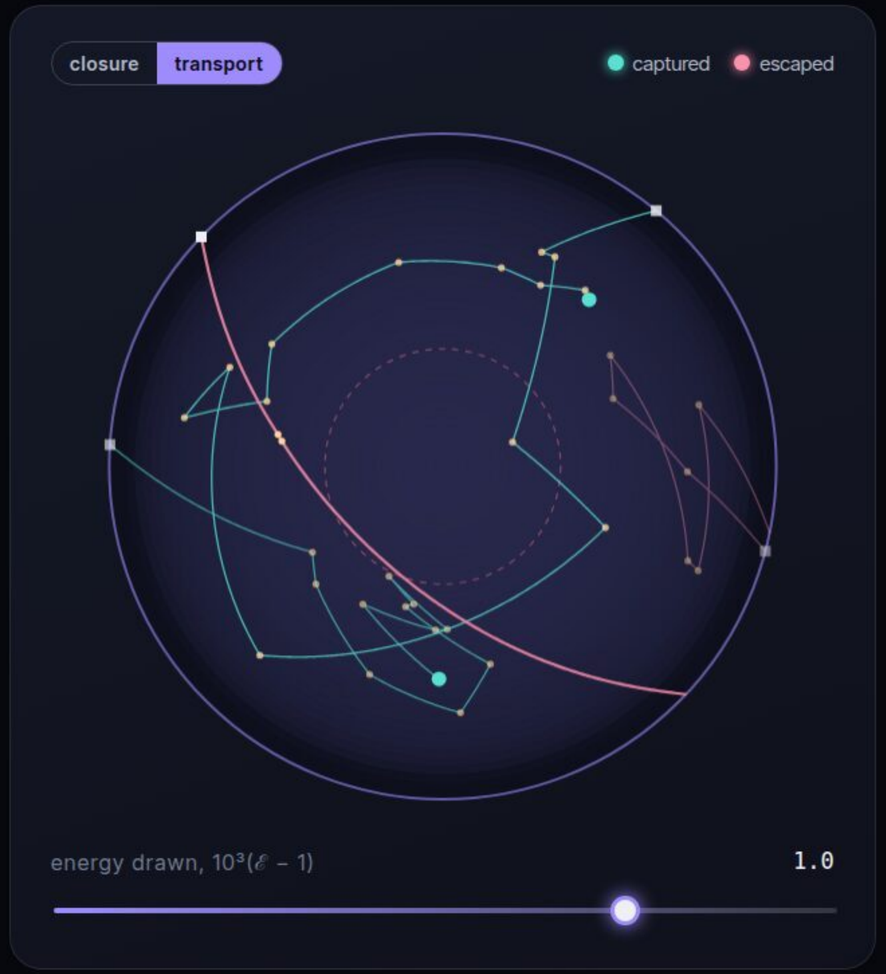
**EVERY APPROXIMATION HAS A NAME**

`GEO-8` interior tetrad transport · `COL-3` relativistic event · `PAULI-1` blocked rate · `STAT-1` Wilson intervals — each documented, cited in the source, and switchable.

Switch off the release, the sampling and the local lapse, and the Monte Carlo **is** the closure.

THREE MODES, ONE STAR · LIVE

# Releasing the trajectory *broadens* the transition






REDUCTION

# Switch it off, and it *is* the closure

| SWITCH OFF | RECOVERS |
|---|---|
| trajectory update at collisions | the incident geodesic of $\tau$ |
| sampled energy loss | the recurrence behind $N_{min}$ |
| loss at the collision radius | the cascade at the surface lapse |
| histories beyond one crossing | the single-crossing convention |
| neutrons at rest in the rate | not switched: up to 0.035 in $P_{cap}$ |

**$N_{min}$**

every capture of the realised closure happens at exactly $N_{min}$

**$0.2_{\%}$**

mean count vs $\sigma\int n_b dl$ by an independent quadrature

**$2.8\sigma$**

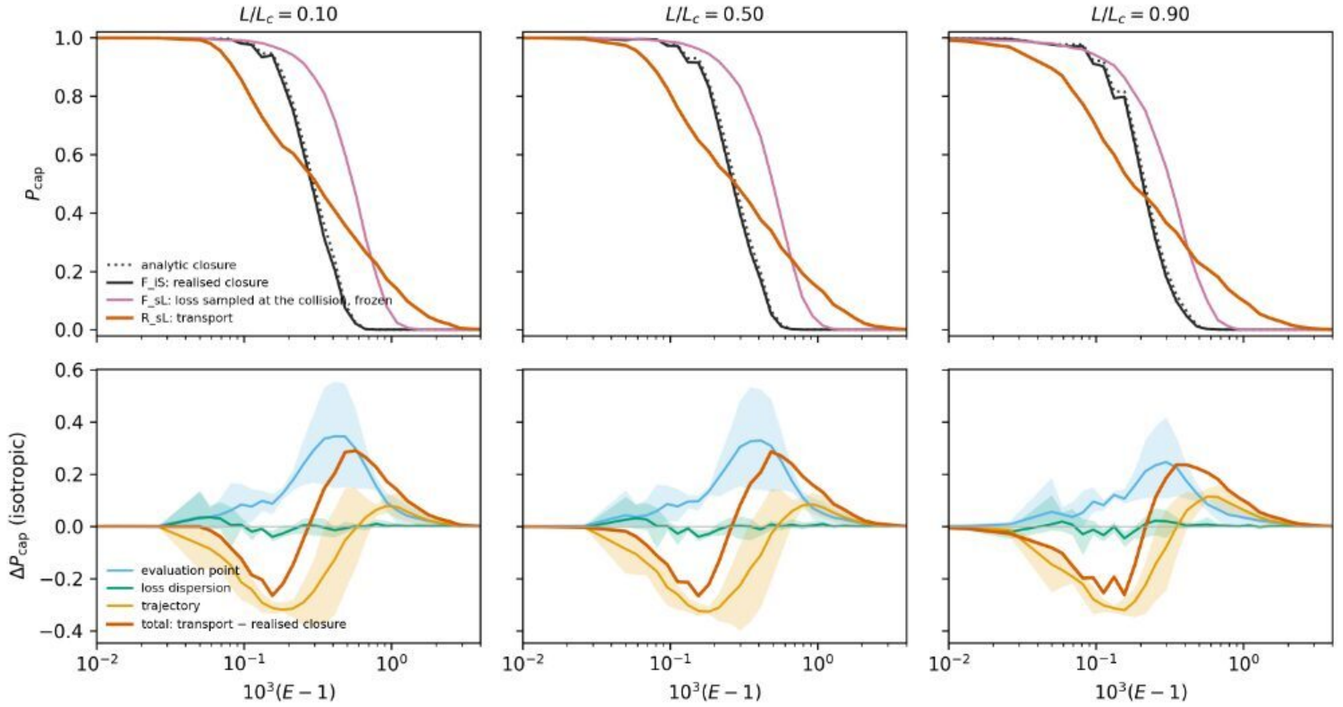
largest deviation of $P_{cap}$ from the Poisson tail, 123 points × 5000 histories

**$10^{-10}$**

a sampled event vs an independent closed form; head-on = one step of the recurrence

THE DECOMPOSITION · $2^3 \times 2$ CONTROLS, ORDER-AVERAGED

# Two effects of *opposite sign*




| SWITCH | MEAN $\Delta P_{CAP}$ |
|---|---|
| • evaluation point | +0.09 … +0.14 |
| • loss dispersion | $\lvert\cdot\rvert$ < 0.01 |
| • trajectory | −0.09 … −0.11 |
| total (signed mean) | +0.00 … +0.03 |
| total, RMS over the transition | 0.16 … 0.17 |

They **nearly cancel in the mean** but not in shape: the first shifts the transition up, the second broadens it. Single energies differ by up to **±0.29**.

An attribution with interactions, not three independent corrections.

Benchmark star, 100 keV, $\sigma = 10^{-44}$ cm$^2$, 5000 histories per point, no blocking.

# Half the particles *turn back*

**48–53%**

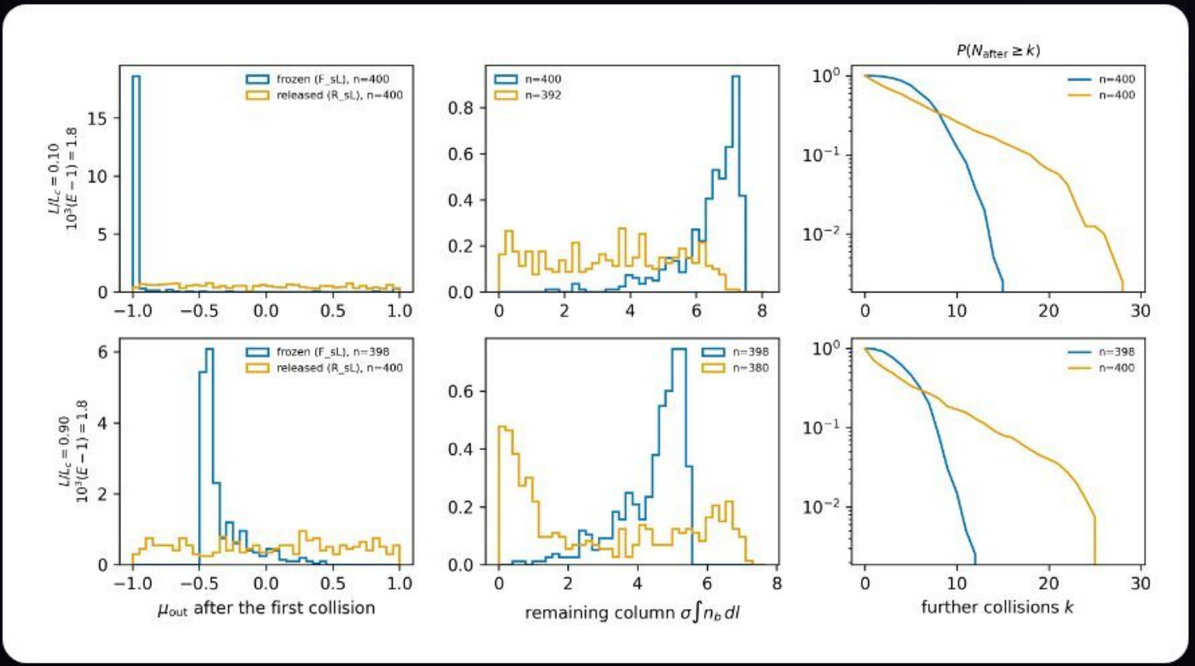
leave the first collision moving **outward** — frozen chord: 2–8%

**≈ 3.1**

mean column left to the surface, spread over 0.2–6.6 (10–90%) — frozen: 4.4–6.5, narrow

**16–28%**

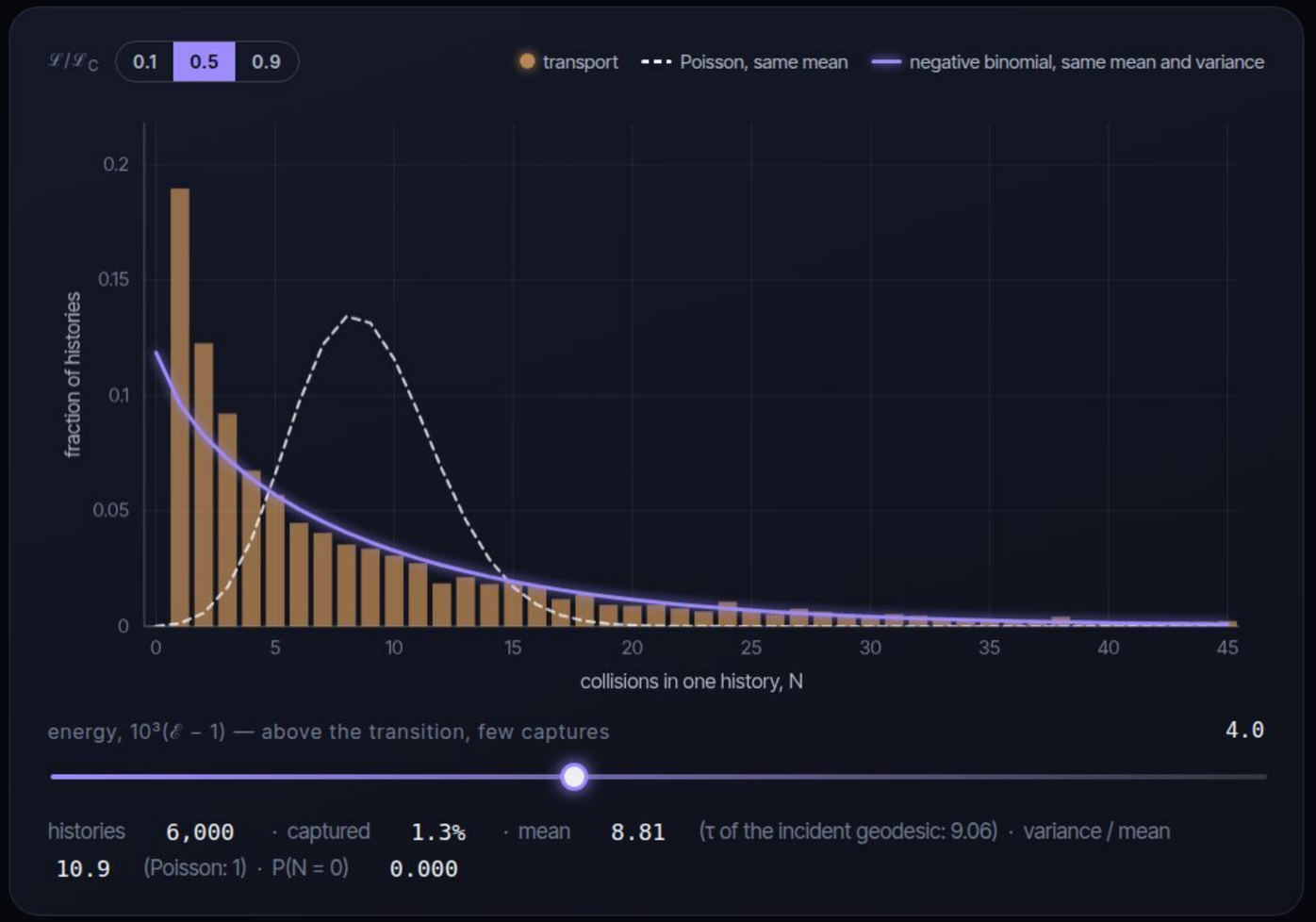
never collide again — frozen: at most 3%




Frozen (F_sL) vs released (R_sL) after the first collision, at $10^3(\mathscr{E} - 1) \approx 1.8$, for $\mathscr{L}/\mathscr{L}_c = 0.1$ and 0.9.

HOW MANY COLLISIONS · LIVE

# A count that is *not Poisson*



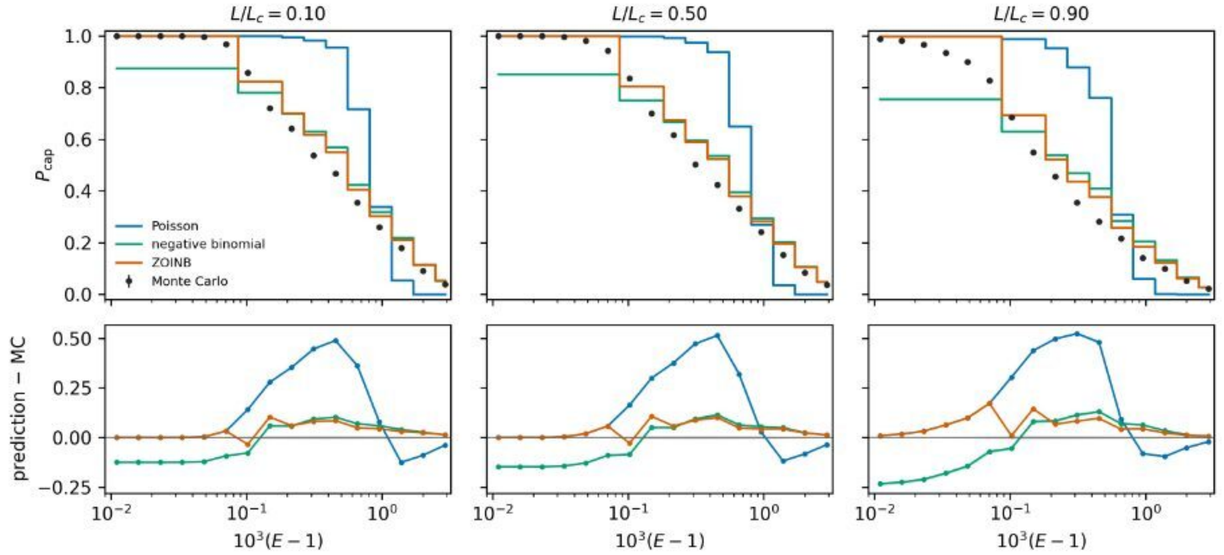

Above the transition almost nothing is captured, so each count is complete: the history ends when the particle leaves. The closure assumes Poisson with the mean τ. The **mean** is right; the **shape** is not.

$$P_{\mathrm{NB}}(k; r, p),\ r = \frac{m^2}{v - m}, p = \frac{m}{v}$$

POISSON COUNT WITH A GAMMA-DISTRIBUTED T

In the paper: variance **8–13 times** the mean. This is a random walk out of an optically thick sphere, not a straight chord.

# Better laws, *honest residuals*




| MEAN \|PREDICTION − MC\| | OUR SLY | BENCHMARK |
|---|---|---|
| Poisson, measured mean | 0.15–0.19 | 0.16–0.18 |
| negative binomial | 0.08–0.11 | 0.057–0.064 |
| zero-and-one-inflated NB | 0.035–0.057 | 0.044–0.064 |

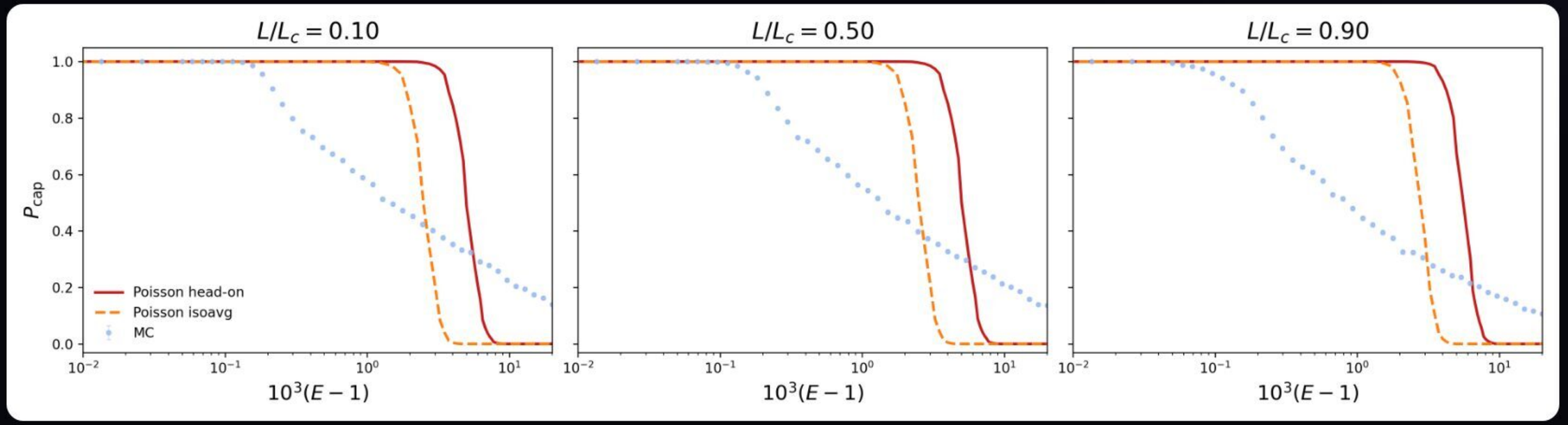
Calibration and test share **no histories**. The zero-and-one-inflated form removes the spurious zero-collision mass of the plain NB.

Residuals stay 6–23 times the combined Monte Carlo and calibration errors: closed forms that help, tested at fixed parameters — not yet a transferable law.

SLy 1.40 M☉, 11.70 km; calibrated above the transition on independent runs, tested inside it.

A MORE COMPACT STAR · SLY AT MAXIMUM MASS, 2.05 $M_\odot$

# Broadening is *not a detail*




Points: the transport. Red: closure with head-on $N_{min}$; orange dashed: isotropic-mean $N_{min}$. 100 keV, $\sigma = 10^{-44}$ cm², no blocking.

Optical depth of the closure: $\tau \approx$ **25–27**, three times the canonical star.

The transport already falls below 0.95 at $10^3(\mathscr{E} - 1) = 0.1$–$0.2$, and keeps **11–14%** at 20, where both closures vanish.

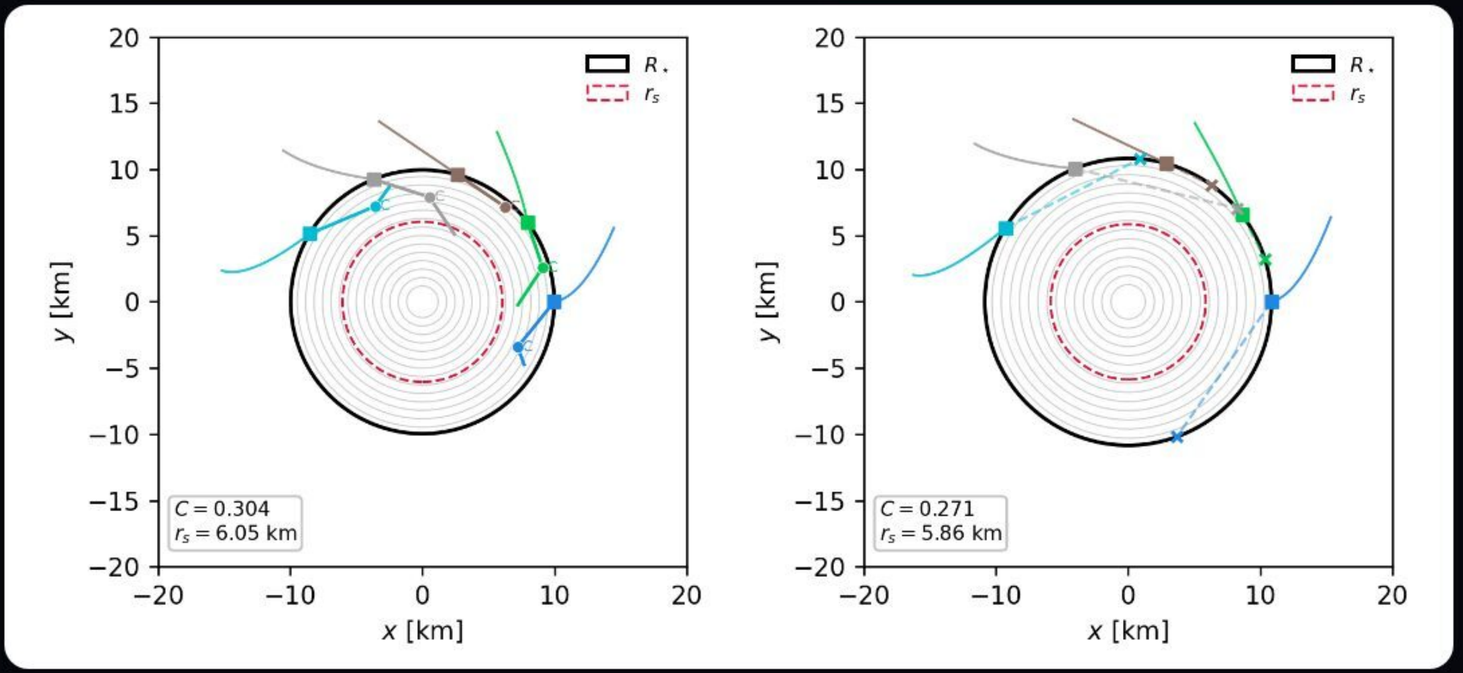
Mean |difference| from the isotropic-mean closure: **0.28–0.29**; single energies up to **0.6–0.7**.

SAME FIVE PARTICLES · DIFFERENT STARS

# The equation of state *decides the fate*

SLY · 2.05 M⊙, 9.96 KM · 5/5 CAPTURED

SQM3 · 1.98 M⊙, 10.84 KM · 0/5 CAPTURED




Identical $\mathscr{E}$, $\mathscr{L}$, speed and impact parameter in both panels; σ = $10^{-44}$ cm$^2$. Squares: entry; dots: collisions; crosses: escape.

Five incident particles, the **same** phase-space points, sent into the maximum-compactness stars of a hadronic (SLy) and a strange-quark (SQM3) equation of state.

The same orbit crosses a different column and collides at a different depth: **five captures against none**.

Selected cases, not an ensemble average — an illustration of where the EOS dependence of $P_{cap}$ comes from.

BEYOND THE BENCHMARK

# Pauli blocking and *cooling*

### PAULI BLOCKING · IMPLEMENTED, NOT YET IN THE CAMPAIGNS

The recoiling neutron must leave the Fermi sea: the blocked optical depth is **$7\times10^{-5}$** of the unblocked one.

But the collisions that do happen are violent: on average **57%** of the kinetic energy, against $2\times10^{-4}$ for a static target.

So a blocked $\tau$ can never be paired with a static $N_{min}$: both must come from the **same blocked kernel**. The hierarchy has to be measured again, not rescaled.

Rate tabulated in ($\ln n_b$, $\ln(\gamma - 1)$) to $10^{-3}$, the Fermi-surface shell of width $\sim3\times10^{-4}$ resolved by bisection.

### COOLING TO 3KT/2 · AFTER CAPTURE

| | $10^7$ K | $10^8$ K |
|---|---|---|
| collisions to capture | 5 | 5 |
| collisions to the threshold | 4494 | 1374 |
| re-entries into the star | 396 | 191 |
| coordinate time | 5.1 s | 6.8 s |
| time spent outside the star | 97% | 99% |

Medians. A first-passage time in a cooling-only model — no heating, no detailed balance: **not** thermal equilibrium.

UNDER THE HOOD

# Validated *stage by stage*

| CHECK | AGREEMENT |
|---|---|
| interior transport vs independent orbit quadrature (path, azimuth) | $1–2\times10^{-4}$ |
| $P_{cap}$ with the step refined from $R/4096$ to $R/16384$ | $\le 2\times10^{-4}$ |
| exterior periapsis vs analytic turning point | $< 1.4\times10^{-13}$ |
| GR / Newtonian focusing at $v \to 0$ | 1.244 |
| surface hits: quadrant asymmetry, $10^6$ samples | $3\times10^{-3}$ |
| CPU vs CUDA, 60 cases × $10^5$ histories | $4.5\times10^{-3}$ |
| population equation vs $N = C_\chi t$ | $10^{-16}$ |
| Fermi EOS: $\Gamma_{eff}$ at the two ends | $1.6667 \cdot 1.3333$ |




**3**

independent implementations of the $N_{min}$ recurrence, same integers

**425**

assertions in 26 test suites

C++17 with OpenMP; the same engine compiled to WebAssembly for an interactive lab.

# What the closure *misses*

- The Monte Carlo reduces **exactly** to the Poisson closure when its released ingredients are switched off.
- Where the loss is evaluated and whether the trajectory is released move $P_{cap}$ by **±0.1** each, in opposite directions: right on average, **0.16–0.18 RMS** off in shape.
- Collision counts are strongly **overdispersed**; negative-binomial laws calibrated out of sample do much better than Poisson.
- Pauli blocking changes rate **and** kernel together: it must be measured, not rescaled.

> *A correct mean number of collisions is not enough for a Poisson closure.*
>
> the one-line summary of Part II

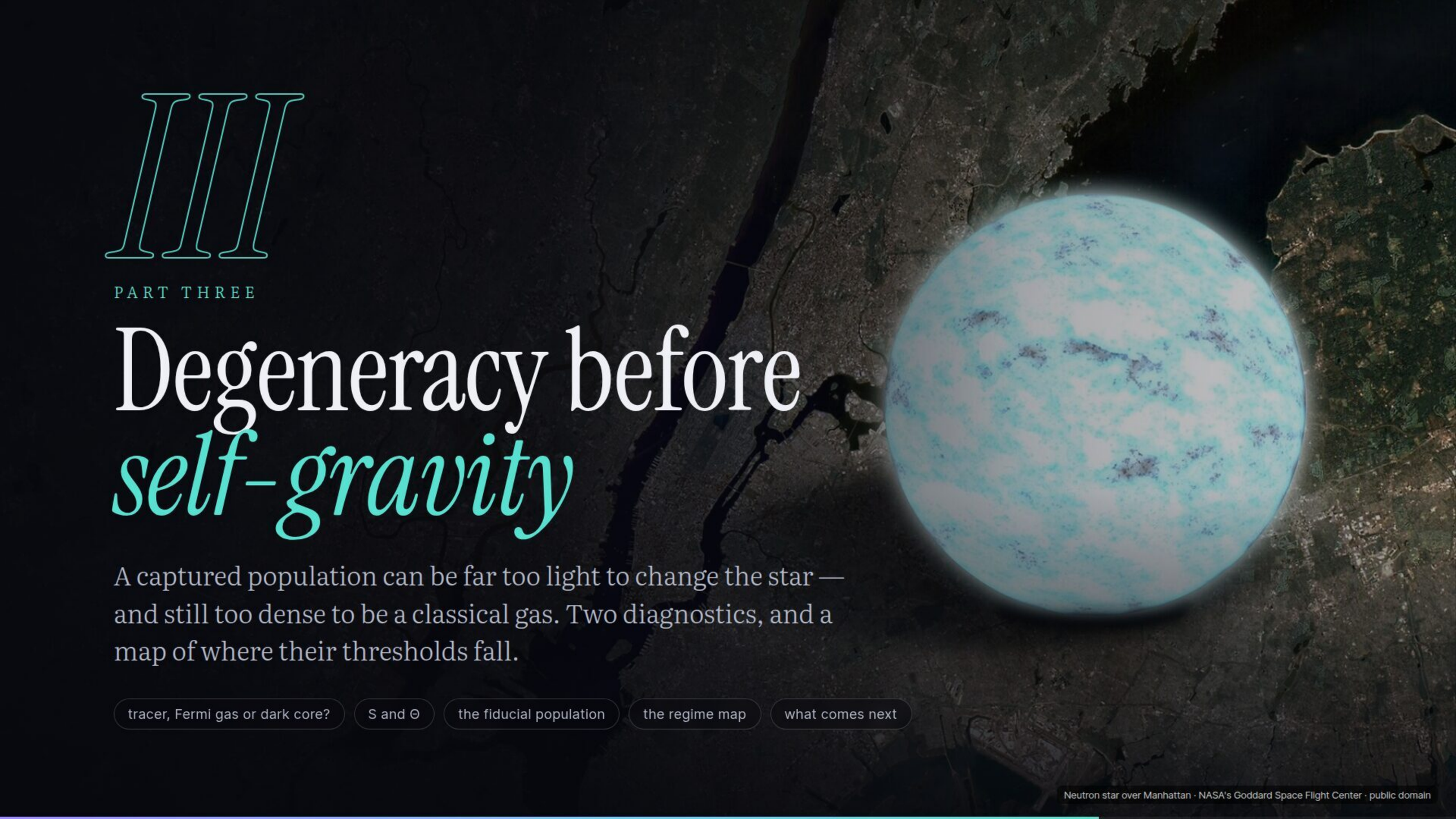
III

PART THREE

# Degeneracy before *self-gravity*

A captured population can be far too light to change the star — and still too dense to be a classical gas. Two diagnostics, and a map of where their thresholds fall.

tracer, Fermi gas or dark core? | S and Θ | the fiducial population | the regime map | what comes next



Neutron star over Manhattan · NASA's Goddard Space Flight Center · public domain

THE QUESTION

# After capture, *what is it?*

capture supplies $N_\chi(t)$ → **which regime?** → the right next model

### A CLASSICAL TRACER

The usual assumption: the particles thermalise into a Boltzmann cloud in the fixed potential of the star. Right for a dilute gas.

$$n \propto e^{-m\Delta\Phi/kT}$$

### A DEGENERATE TRACER

Fermions in a deep well can reach a phase-space density where **Pauli** matters — while still too light to change the gravity.

$$E_F \gtrsim kT$$

### A DARK COMPONENT

Dark-matter admixed neutron stars: the dark fluid takes part in hydrostatic balance and shifts the mass–radius relation.

two-fluid TOV

TWO INDEPENDENT DIAGNOSTICS

# One for gravity, *one for statistics*

$$S = \frac{M_\chi(r_\chi)}{M_b(r_\chi)}$$

SELF-GRAVITY · INSIDE THE REGION THE DARK MATTER OCCUPIES

| S | REGIME | OMITTED GRAVITY S/(1 + S) |
|---|---|---|
| $< 10^{-3}$ | tracer | $< 10^{-3}$ |
| $10^{-3} – 10^{-1}$ | transitional | up to 9% |
| $\geq 10^{-1}$ | self-gravitating | ≥ 9% |

Local, not the global $M_\chi/M_\star$: $r_\chi = r_{90}$ holds 90% of the particles.

$$\Theta = \frac{T}{T_F} = \frac{k_B T}{E_F}$$

CLASSICALITY · THERMAL ENERGY OVER FERMI ENERGY

| Θ | REGIME | FERMI SHARE 1/(1 + Θ) |
|---|---|---|
| > 10 | thermal | < 0.09 |
| 0.1 – 10 | transitional | 0.09 – 0.91 |
| < 0.1 | degenerate | > 0.91 |

There is no general reason for the two sets of thresholds to **coincide**.

THE TWO CLOSURES

# A Boltzmann cloud, and *the Fermi limit*

$$n_\chi(r) = n_0 \exp\left[-\frac{m_\chi \Delta\Phi(r)}{k_B T}\right]$$

THERMAL TRACER IN THE FIXED POTENTIAL OF THE STAR

$$N_\chi = 4\pi \int_0^R n_\chi r^2 \sqrt{g_{rr}}\, dr$$

FIXES $n_0$ · $r_{90}$ HOLDS 0.9 $N_\chi$

$$p_F = \hbar\,(3\pi^2 n_\chi)^{1/3},\ x_F = \frac{p_F}{m_\chi c}$$

SPIN ½, g = 2

$$E_F = m_\chi c^2\,(\sqrt{(1 + x_F^2)} - 1)$$

$P \propto \rho^{5/3}$ for $x_F \ll 1$ · $P \propto \rho^{4/3}$ for $x_F \gg 1$

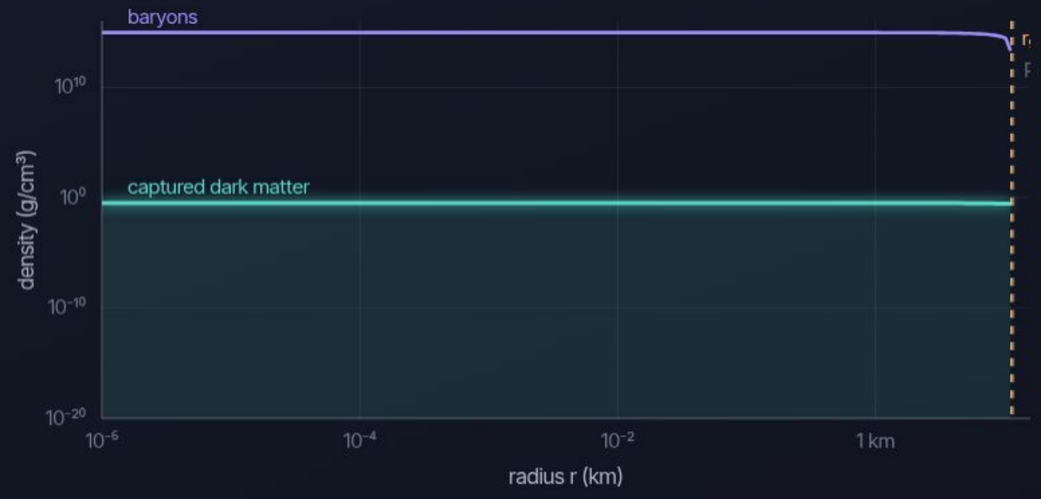
The cloud has no edge: $r_{90}$ ignores the dilute tail and measures where the particles actually are.

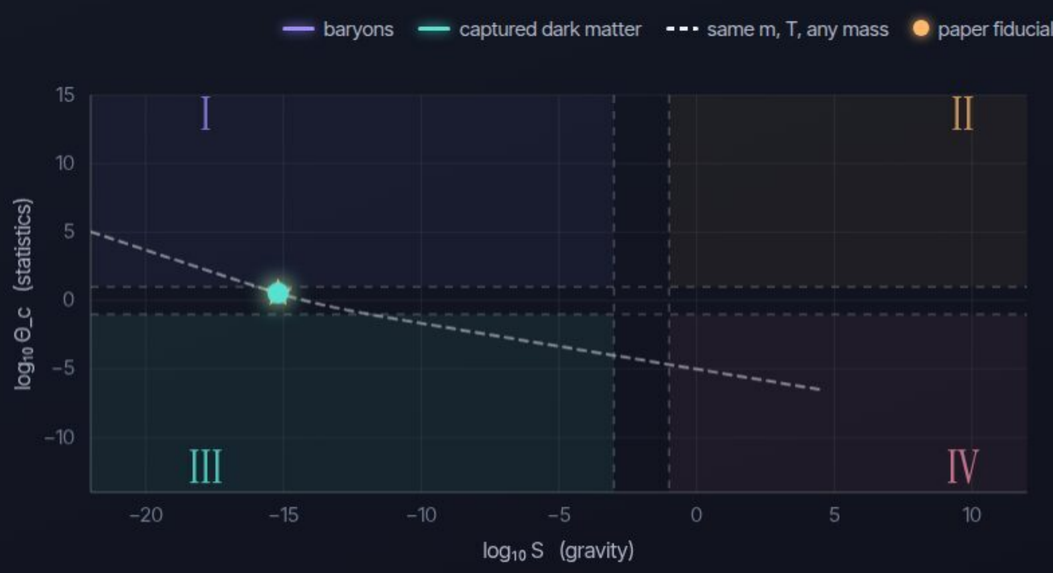
$\Theta$ is a **diagnostic ratio**, not a finite-temperature chemical potential: it says when a Fermi–Dirac treatment is needed.

THE TWO DIAGNOSTICS · LIVE

# Move a cloud through *the plane*

PRESETS | paper fiducial · 100 keV, $10^9$ K | 1 GeV WIMP, $10^5$ K | 1 keV, $10^6$ K | $10^6$ GeV, $10^5$ K




Dashed lines: the thresholds $S = 10^{-3}, 10^{-1}$ and $\Theta_c = 10, 0.1$. The bands between them are transitional.

*I* Boltzmann tracer $S < 10^{-3}$, $\Theta_c > 10$

*II* Boltzmann, self-gravitating $S \geq 10^{-1}$, $\Theta_c > 10$

*III* Fermi tracer $S < 10^{-3}$, $\Theta_c < 0.1$

*IV* Fermi, self-gravitating $S \geq 10^{-1}$, $\Theta_c < 0.1$

particle mass $m_\chi$ 100 keV

core temperature 1.0×10⁹ K

captured mass $M_\chi$ 2.36×10¹⁸ g

$r_{90}$ 11.3 km · central density 0.328 g/cm³ · x_F 0.748 · E_F 24.9 keV vs kT 86.2 keV

$S$ 6.37×10⁻¹⁶ TRACER · Θ_c 3.46 TRANSITIONAL

· on this line Θ_c = 1 at $S \approx 10^{-14}$

# Pauli comes *first* for light particles

A compact cloud sits in the harmonic core, $\Delta\Phi = (2\pi/3)G\rho_c r^2$. Its size cancels between $S$ and the central density:

$$n_0 \approx 4.6\, S\, \frac{\rho_c}{m_\chi}$$

INDEPENDENT OF T

$$\Theta_c \propto T\, m_\chi^{5/2}\, S^{-2/3} \Rightarrow S(\Theta_c = 1) \propto \frac{m_\chi^{5/2}\, T^{3/2}}{\rho_c}$$

NON-RELATIVISTIC FERMI GAS

Light particles are many per gram: they fill phase space long before they weigh anything.

| PARTICLE, CORE | S AT $\Theta_c = 1$ |
|---|---|
| 1 keV, $10^6$ K | $10^{-24}$ |
| 100 keV, $10^9$ K | $10^{-14}$ |
| 1 GeV, $10^5$ K | $10^{-11}$ |
| 1 GeV, $10^7$ K | $10^{-8}$ |
| 1 TeV, $10^5$ K | $10^{-4}$ |

Self-gravity starts at $\mathbf{S \sim 10^{-3}}$. For anything lighter than about a TeV, the gas turns **degenerate first**.

Full thermal profile in the SLy star, as in the previous panel.

THE PIPELINE

# From the star to *the diagnostics*

baryonic TOV star → $b_{max}(V)$ → $\Gamma_{inc}$ → $f_{cap}$ (Monte Carlo) → $C_\chi$ → $N_\chi(t)$ → $\rho_\chi(r)$ → (S, Θ)

$$\Gamma_{inc} = n_\chi \pi \int b_{max}^2 v\, f(v)\, dv$$

FOCUSED FLUX ONTO THE SURFACE

$$C_\chi = \Gamma_{inc}\, f_{cap}$$

$f_{cap}$: CAPTURED OVER SAMPLED

$$\frac{dN_\chi}{dt} = C_\chi - C_A N_\chi^2 - C_E N_\chi$$

ANNIHILATION, EVAPORATION: OFF IN THE FIDUCIAL

The order matters: capture supplies the particle budget, S tests whether the fixed gravitational background is adequate, Θ tests whether the classical closure is.

# Gravitationally invisible, *not quite classical*

| | MAXWELL–BOLTZMANN | RAR–FD |
|---|---|---|
| incidence rate $\Gamma_{inc}$ | $4.36\times10^{29}$ s$^{-1}$ | $2.33\times10^{29}$ s$^{-1}$ |
| capture fraction $f_{cap}$ | 0.962 | 0.959 |
| captured particles $N_\chi$ | $1.32\times10^{46}$ | $7.06\times10^{45}$ |
| captured mass $M_\chi$ | $2.36\times10^{18}$ g | $1.26\times10^{18}$ g |
| cloud radius | 11.3 km ($r_{90}$) | 11.7 km (support) |
| self-gravity $S$ | $6.4\times10^{-16}$ | $3.8\times10^{-16}$ |
| classicality $\Theta_c$ | 3.46 | 5.07 |

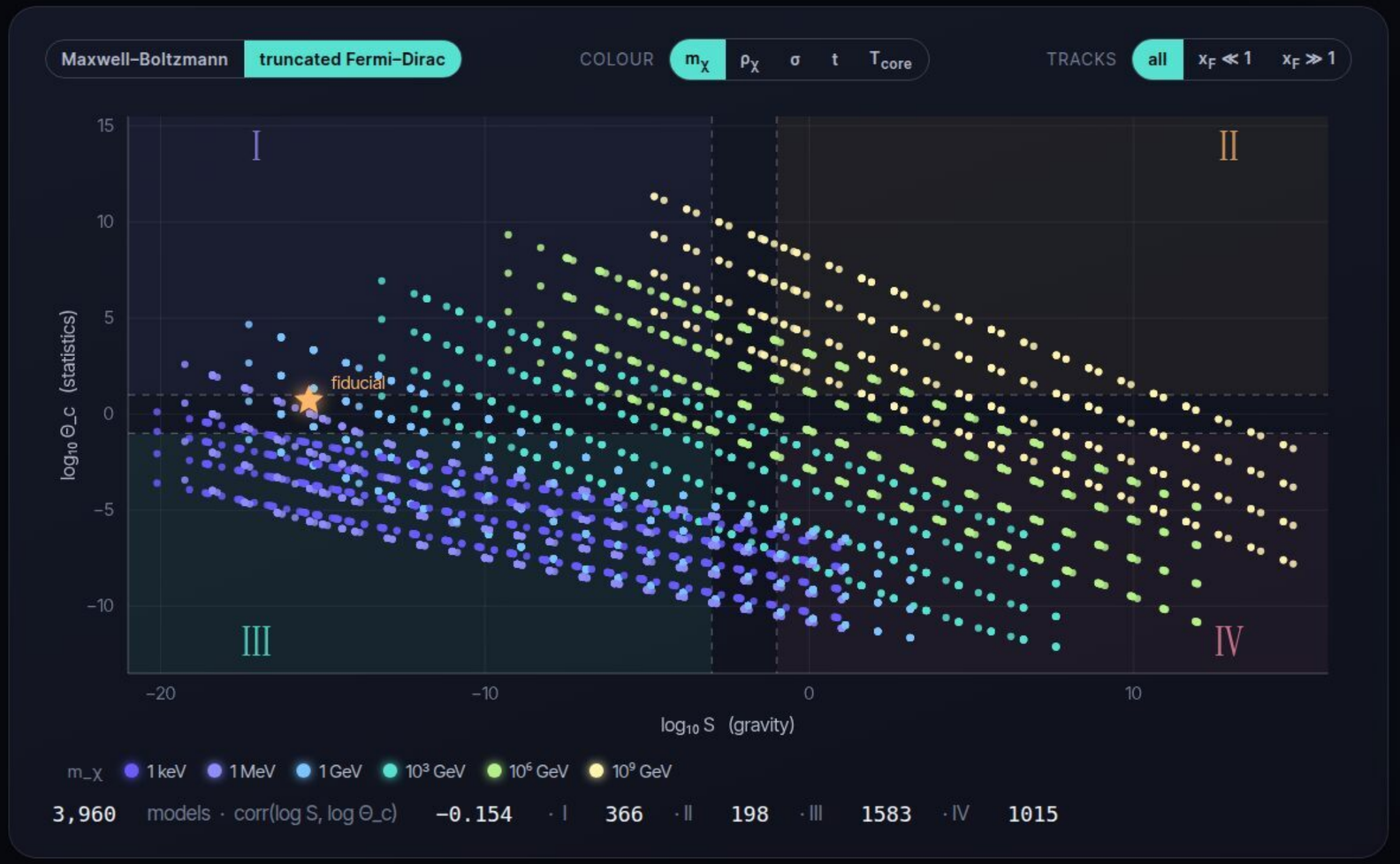
**Fiducial:** $m_\chi = 100$ keV, $\sigma = 10^{-44}$ cm$^2$, $\rho_\chi = 0.4$ GeV cm$^{-3}$, $T_{core} = 10^9$ K, $t = 10^9$ yr, SLy 1.40 $M_\odot$.

$M_\chi/M_\star \approx 10^{-15}$: no mass–radius shift to look for. Yet $\Theta_c \approx 3$–5 is in the **transitional** band: corrections of order one to Boltzmann.

Changing the halo from MB to RAR–FD changes the **supply** by 47% and the microscopic capture by 0.3%.

THE REGIME MAP · EVERY MODEL OF THE SURVEY

# The two boundaries *do not coincide*




## *I*

**BOLTZMANN TRACER**

the usual assumption

## *II*

**BOLTZMANN, SELF-GRAVITATING**

heavy, classical, massive

## *III*

**FERMI TRACER**

degenerate, gravitationally passive

## *IV*

**FERMI, SELF-GRAVITATING**

toward a dark core

READING THE MAP

# Two slopes, *one Fermi energy*

Hold $m_\chi$, $\sigma$, $t$, $T$ fixed and change $\rho_\chi$: the profile keeps its shape, $S \propto n_0 \propto N_\chi$, and $p_F \propto n_0^{1/3}$:

$$\Theta_c \propto S^{-2/3}$$

$x_F \ll 1 \cdot E_F \approx p_F^2/2m$

$$\Theta_c \propto S^{-1/3}$$

$x_F \gg 1 \cdot E_F \approx p_F c$

| RAR–FD SURVEY | TRACKS | POINTS | MEDIAN SLOPE |
|---|---|---|---|
| non-relativistic family | 352 | 2112 | −0.667 |
| relativistic family | 213 | 1278 | −0.334 |

### THE MASS TREND

Heavier particles are less degenerate: $d \log \Theta_c / d \log m_\chi \approx$ **0.49** (MB) and **0.42** (RAR–FD). The degenerate quadrants hold **85–89%** of the lightest models and **21–28%** of the heaviest.

### NOT A CHANGE OF PARTICLE

III → IV is the same Fermi gas becoming heavier where it sits: the growth of $S$, not of the particle mass.

MB vs RAR–FD: corr($\log S$, $\log \Theta_c$) = −0.652 vs −0.154.

WHAT COMES NEXT

# The right model *for each quadrant*

### *I*

$S \ll 1,\ \Theta \gg 1$

a Boltzmann cloud in the fixed star — the standard assumption, justified

### *II*

$S \gtrsim 0.1,\ \Theta \gg 1$

classical, but it shapes its own potential

### *III*

$S \ll 1,\ \Theta \lesssim 1$

a capture-normalised **Fermi tracer** in the fixed star — finite-T if $\Theta \sim 1$–10

### *IV*

$S \gtrsim 0.1,\ \Theta \ll 1$

only here: a **two-fluid TOV**, the dark-matter admixed star

Capture calculations and admixed-star models are not interchangeable endpoints. A small $M_\chi/M_\star$ or $S$ says the structure is unchanged; it does **not** justify a Boltzmann profile.

**NOT YET INCLUDED**

Annihilation heating, collapse and black-hole formation, rotation, magnetic fields, and a self-consistent time evolution of the captured Fermi gas. $\Theta$ is a diagnostic, not a finite-temperature solution.

CONCLUSIONS

# Capture, *then character*

- **OBSCURA** follows dark-matter collision histories through the interior metric and reduces exactly to the optical-depth closure.
- At 100 keV, the frozen trajectory and the surface evaluation each move $P_{cap}$ by **~0.1**, in opposite directions; the count of collisions is **overdispersed**.
- After capture, **S** and **Θ** answer different questions: the fiducial population is an extreme tracer ($S \sim 10^{-16}$) in the transitional band ($\Theta_c \approx 3$–$5$).
- For light particles **degeneracy comes before self-gravity**: the thermalised-WIMP assumption has to be tested in (S, Θ), not by $M_\chi/M_\star$.

**NEXT STEPS**

- Pauli-blocked capture campaigns with the consistent blocked kernel.
- Capture rates over the full incident phase space, RAR halo included.
- Thermalisation with heating and detailed balance, then a finite-T Fermi tracer.
- Two-fluid structure once S leaves the tracer regime.

The panels in this seminar run in the browser, on the same physics as OBSCURA: *rafael-lima.pages.dev/talks*.

IMAGE CREDITS

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**Paper figures** — de Lima & Rueda, *A Monte Carlo relativistic framework for dark matter capture by neutron stars* (in preparation); de Lima & Rueda, *Degeneracy Before Self-Gravity: Dark Matter Regimes in Neutron Stars* (draft)

**Data in Part I** — RAR model: solved in the browser after Argüelles et al., PDU 21, 82 (2018); rotation curve: SPARC (Lelli, McGaugh & Schombert 2016); CMB: Planck 2018 and CLASS v3.4.1 (Blas, Lesgourgues & Tram 2011); direct-detection and axion limits: compilations by C. O'Hare (DirectDetectionPlots; AxionLimits, Zenodo 3932430), neutrino fog: O'Hare 2021

**Regime-map panel** — the survey data of the second paper, from the OBSCURA pipeline

**Neutron star** — SLy equation of state (Douchin & Haensel 2001) in the piecewise-polytropic fit of Read, Lackey, Owen & Friedman, PRD 79, 124032 (2009)

**OBSCURA logo** — OBSCURA project

THANK YOU

# *Questions?*

Slides and live panels, on the same physics as OBSCURA, at
*rafael-lima.pages.dev/talks*


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